KEYWORDS: Chatbot, Retrieval-Augmented Generation, Hybrid Retrieval, Academic Advising, Hallucination Detection, Code-Mixed NLP, Natural Language Processing, Educational Technology, Information Retrieval.

Abstract: This paper's contribution is at the mechanism and methodology level, not the ranking-formula level, and we lead with the two results that make that case most directly. First, an ambiguous-entity disambiguation mechanism (`conditioning_v2`) that, when a named entity resolves to more than one real record, builds its clarifying question directly around a corpus field the tied candidates actually disagree on, rather than asking the user to simply retype a full name – validated at $n = 220$ against both a plain notice and an earlier, less directive version of the same idea, significantly better on all three judged criteria ($p < 0.0001$), and, to our knowledge, a genuine differentiator against the closest related system, BanglAssist (Kruk et al., 2025), whose only confidence-adjacent component is a fixed cosine-similarity threshold with no entity-disambiguation mechanism at all (Section 2). Second, a deconfounding-diagnostic methodology for separating a routing decision's own contribution from a masking downstream mechanism: isolating adaptive fusion routing's own method choice from a separate exact-match ranking guarantee found the original choice (Reciprocal Rank Fusion for entity-heavy queries) was significantly negative on entity-heavy retrieval once that guarantee was removed, not merely neutral as a naive pooled comparison had suggested – the guarantee was masking a real regression, not confirming the routing choice was sound. We acted on this rather than only disclose it: replaced RRF with linear fusion at the same weighting for entity-heavy queries, verified live that this changes no observed behavior on the current test set (the exact-match mechanism already determines every real outcome), and measured a genuine, confirmed improvement in the residual ranking-quality gap as a direct result.

These two mechanisms sit inside a broader adaptive pipeline, evaluated end-to-end on 200 English and 100 Banglish (Bengali-English code-mixed) test queries with paired significance testing and standard IR metrics (Recall@k, MRR, nDCG@10) throughout: domain-fine-tuned bilingual sentence embeddings trained via contrastive (question, answer) pairs on a strict train/validation/test split; per-query-type fusion routing (entity-heavy queries through a heavily lexical-weighted blend, open-ended queries through a more balanced one); prerequisite-graph augmentation, answering multi-hop prerequisite-chain questions by traversing a directed graph built from the knowledge base's own edge data, a query shape neither lexical nor dense retrieval can answer from a single chunk; and confidence-gated abstention, calibrated per route against a labeled out-of-scope subset (augmented with database-verified synthetic examples), with a sufficient-context override that prevents the system from declining to answer when direct structural evidence is already in hand.

As supporting evidence that these two mechanisms and the pipeline around them come at no measurable cost – not as this paper's central claim – we report every automated-metric result obtained, including negative ones. A third mechanism, conformal claim-level backoff (Mohri and Hashimoto, 2024), was implemented but never calibrated in prior work; we calibrate it for the first time against a real, structurally-labeled 442-claim set with a proper held-out split, find it degenerates to a threshold that empty-filters every held-out claim (0/133 retained), and diagnose the precise cause – 88% of the calibration set's negative-labeled claims are near-verbatim reproductions of their own (irrelevant) retrieved context, so NLI entailment scores them as faithful despite the answer being wrong, a structural mismatch no recalibration can fix – rather than leave the mechanism in undecided limbo (Section 4.10.2). A cross-encoder reranker – both generic and, after we caught and corrected a training/deployment mismatch, properly fine-tuned on the corpus's own hard negatives – was found to significantly *hurt* the pipeline in both cases and is disabled by default; BM25-level Banglish spelling-variant normalization shows a null effect on the main Banglish test set. Cross-lingual query translation also showed a null effect on that same test set, but we traced this to a structural blind spot in the test set itself and found translation still helps (61.5% to 84.6% top-5 accuracy, $n = 13$, a real replication of the same direction at a still-small but growing sample) on a dedicated stress test built for the scenario it targets, though a larger, noisier expanded version of that same stress test no longer reaches even suggestive statistical support after this revision's other retriever improvements partly closed the gap it was designed to detect. Under the current retriever and corpus (hard-negative-mined embeddings, an exact-match coverage fix, a BM25 k_1/b retuning, a paraphrase-robustness fix, and a prerequisite-chain corpus/vector-index fix), the full adaptive pipeline is in full statistical parity with every single-method baseline on all four automated similarity metrics: tied with full hybrid and BM25-only on all four ($p \geq 0.530$ throughout), and against vector-only the closest case, METEOR, is significant under a Wilcoxon test ($p = 0.018$) but not under a paired t-test ($p = 0.128$), so we treat it as inconclusive rather than confirmed, consistent with how we treat other test-disagreeing results throughout this paper. Parity is not automatically an advantage on its own, which is exactly why the deconfounding diagnostic above matters: it is what let us tell apart genuine parity from parity borrowed from a separate mechanism. The system's demonstrated value beyond the two headline mechanisms is therefore in capabilities the single-method baselines cannot perform at all, not in beating them on automated similarity metrics: abstaining on only 0.5% of English and 2.0% of Banglish queries it cannot ground (down from 10.5% before a coverage fix to the exact-match mechanism), correctly answering multi-hop prerequisite-chain questions via graph traversal, and operating over a bilingual corpus with comparable results in both languages. An automated LLM-judge faithfulness check (RAGAS-style), re-measured against the corrected corpus, descriptively ranks the adaptive pipeline highest of four configurations (0.861 vs. 0.825–0.853 for the three retrieval baselines); a matched-subset paired test finds this raw ordering is not itself confirmed – every comparison against a real retrieval baseline (BM25-only, full hybrid, vector-

only) resolves to a non-significant tie, significant only against no-retrieval – and a second, architecturally independent NLI-based faithfulness check shows only weak, and on re-measurement more strongly chance-level, agreement with the LLM judge at the row level, the two instruments now disagreeing even on which configuration scores best. Together, this is evidence too inconclusive to confirm a genuine faithfulness advantage or regression in either direction, which we report as an open question rather than resolve based on whichever raw number a given measurement round happens to produce. The underlying retrieval mathematics (lexical/dense linear fusion, RRF) is established, not novel, and every generation-quality comparison against single-method baselines lands at parity rather than a win; a bilingual application of sufficient-context classification, not reported elsewhere in the literature we surveyed, rounds out this paper's mechanism-level contributions alongside the two above. We disclose every limitation and methodological pitfall we found along the way, including our own.

Adaptive Hybrid Retrieval with Confidence-Gated Abstention for Bilingual Academic Advising

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1 INTRODUCTION

Open-credit university systems place substantial demands on students to independently plan course sequences, interpret evolving policy documents, and schedule their studies, while mentor availability per student remains limited. Despite the growing availability of digital academic resources, students at open-credit institutions continue to report difficulty accessing timely, personalized guidance on course selection and policy questions, particularly during the first year (Nguyen et al., 2023). This gap is especially acute for first-year students, who are simultaneously adjusting to a new academic environment and to the self-directed nature of open-credit planning. At the institution this system targets, a substantial share of student questions are also not posed in formal English but in Banglish – Bengali-English code-mixed text typed in Latin script – a register general-purpose and even most retrieval-augmented university chatbots are not built or evaluated for.

Conversational agents are a natural candidate for addressing this gap: they can process natural language input and sustain multi-turn, human-like conversational exchanges with users (Brandtzaeg and Følstad, 2017), and can offer fast, personalized assistance at a scale that individual mentors cannot (Shawar and Atwell, 2007). However, general-purpose conversational agents built on large language models (LLMs) are prone to hallucination – the generation of fluent but factually incorrect content – which is particularly risky in an advising context where a single incorrect prerequisite or deadline can have real consequences for a student's academic trajectory.

This paper substantially extends our own earlier system, Rahman et al. (2025), a hybrid BM25/ChromaDB academic-advising chatbot for the same institution that established a simple lexical/dense hybrid retriever outperforms either retrieval method alone specifically on entity-heavy queries (alphanumeric course-code identification), while blending in dense retrieval on top of a well-tuned lexical stream neither reliably helps nor hurts on open-ended natural-language questions – using a single fixed fusion weight ($\lambda = 0.5$) throughout. That system's own limitations explicitly identify the absence of Bangla/Banglish support as a gap left to future work, and offer no confidence-gated abstention mechanism, no reasoning over the corpus's prerequisite structure beyond retrieving whichever chunk states it, and no ablation of a reranking stage. This paper's contribution is to take the fixed-fusion-weight finding as a design constraint rather than a conclusion, and to close each of those gaps directly rather than merely name them: if a single global fusion weight cannot be shown to beat lexical retrieval on any query type, the right response is to stop searching for one global weight and instead route different query types differently, add the structured signals (a prerequisite graph, exact metadata matches) that text retrieval alone cannot recover, add a mechanism for declining to answer when none of this is enough, and extend the system to the bilingual setting its predecessor's own limitations called for – rather than accept a single fixed architecture as final. The dataset itself has also been substantially revised for this paper (Section 3.1), not merely re-scored with the predecessor's original data.

The two contributions below are this paper's strongest, most cleanly-differentiated results, and we lead with them rather than with the surrounding pipeline; the rest of this list is the infrastructure that makes them possible and the supporting evidence that adding them costs nothing on standard metrics, not the other way around. Concretely, this paper:

- adds an ambiguous-entity disambiguation mechanism (`conditioning_v2`) that, when a named entity resolves to more than one real record, builds its clarifying question around a field the candidates actually disagree on rather than asking the user to simply retype a full name, validated at $n = 220$ against both a plain notice and an earlier, less directive version of the same idea (all three judged criteria, $p < 0.0001$) – to our knowledge a genuine differentiator against the closest related system, BanglAssist (Section 2), which has no entity-disambiguation mechanism of any kind;

- introduces a deconfounding-diagnostic methodology that separates a routing decision's own contribution from a masking downstream mechanism, used here to find that adaptive routing's original fusion-method choice (Reciprocal Rank Fusion for entity-heavy queries) was significantly negative once a separate exact-match ranking guarantee was isolated out – a regression a naive pooled comparison reported as neutral parity instead – and acted on, not merely disclosed: replaced with linear fusion at the same weighting, verified to change no real-world behavior, and confirmed to narrow the residual ranking-quality gap as a direct result;

- domain-adapts the retriever's sentence embeddings via contrastive fine-tuning on the corpus's own (question, answer) pairs, under a strict split that never trains on validation or test data, and reports the held-out accuracy gain this produces before any other component is touched;

- replaces the single fixed-λ fusion rule with adaptive routing by query type (entity-heavy vs. open-ended), motivated directly by the entity/open-ended asymmetry the prior ablation already established;

- adds a directed prerequisite graph, built from the knowledge base's own edge data, to correctly answer multi-hop prerequisite-chain questions that no single retrieved chunk contains;

- adds a confidence-gated abstention mechanism, calibrated per route against labeled out-of-scope examples, together with a sufficient-context override that prevents abstaining when direct structural evidence for an answer is already available – both signals computed as a side effect of retrieval already performed, at no additional model call;
- extends the corpus and every evaluation to a second language variety, Banglish, and reports the corresponding results and ablations (query translation, code-mixed lexical normalization) with the same rigor, including where they do not help;
- calibrates a conformal claim-level backoff mechanism (Mohri and Hashimoto, 2024) for the first time against a real, structurally-labeled calibration set with a proper held-out split, finds it degenerates to a threshold that filters out every held-out claim, and diagnoses the precise mechanism (88% of negative-labeled calibration claims are near-verbatim reproductions of their own irrelevant retrieved context) rather than leaving a built-but-unvalidated component undecided; and
- reports every ablation's outcome, including the negative ones (a non-fine-tuned cross-encoder reranker, disabled by default) and every methodological limitation found along the way, since the standing requirement for this work is that no reported number be anything other than the output of code that was actually run.

Accordingly, we ask three research questions: (RQ1) does replacing single-method or single-λ retrieval with the adaptive pipeline described above improve automated semantic-quality scores, and if not, what does it add instead; (RQ2) do the individual novel components (fine-tuned embeddings, adaptive routing, graph augmentation, abstention, reranking) each help, hurt, or have no measurable effect, evaluated by direct ablation; and (RQ3) does the system perform comparably on Banglish queries, and do techniques specifically aimed at the code-mixed setting (query translation, lexical normalization) measurably help on this corpus. Sections 2–4 address these questions in turn, and Section 5 discusses their implications together with the system's limitations.

2 RELATED WORK

2.1 Chatbots in Education

Conversational agents have been applied to education since ELIZA, an early system based on pattern matching and scripted responses (Adamopoulou and Moussiades, 2021). Subsequent work has examined chatbots for FAQ-style university support (El-Ashmawi et al., 2023), knowledge-base-driven question answering (Formica et al., 2023), and retention-oriented advising support, such as the Lilo system, which combines advising with social-emotional support to address "summer melt" among newly admitted students (Nguyen et al., 2023). Hwang and Chang (2021) survey the opportunities and open challenges of educational chatbots, noting that evidence of effectiveness across diverse educational settings remains limited. More recently, Swacha and Gracel (2025) survey 47 RAG-based educational chatbots and identify hallucination mitigation as the primary motivation for adopting RAG in this domain, while noting that direct hallucination measurement, rather than similarity-based proxies, remains rare across the surveyed literature. None of the systems surveyed report evaluation on code-mixed or bilingual student input, which is the gap Section 4.11 addresses directly.

2.2 Retrieval-Augmented Generation and Hybrid Retrieval

Retrieval-Augmented Generation (RAG) grounds LLM outputs in externally retrieved evidence rather than relying solely on parametric knowledge, and has become a primary strategy for improving factual reliability in generative systems (Lewis et al., 2020). Sawarkar et al. (2024) demonstrate that blending semantic search with hybrid query-based retrievers improves retrieval accuracy over either method alone, consistent with our design motivation for combining BM25 and dense retrieval where alphanumeric course codes require lexical precision that dense retrieval alone does not reliably provide. Beyond linear score combination, Cormack et al. (2009) introduce Reciprocal Rank Fusion (RRF), a rank-based (rather than score-based) fusion rule that avoids the cross-query scale-comparability problems of linearly combining two differently-distributed similarity scores; Section 3.5 uses RRF specifically for the query type (entity-heavy lookups) where a small number of high-confidence exact matches should dominate the ranking regardless of how the remaining candidates' raw scores happen to be distributed for that particular query.

A growing literature addresses hallucination in RAG-based systems specifically. Huang et al. (2023) provide a taxonomy of hallucination causes and mitigation strategies, situating RAG as one of the most widely adopted inference-time mitigations. Varshney et al. (2023) propose validating low-confidence generations before they reach the user, closely related to this system's confidence-gated abstention (Section 3.7). Joren et al. (2024) show that combining a context-*sufficiency* signal with retrieval/model confidence outperforms confidence alone on selective accuracy-coverage trade-offs, since confidence heuristics such as a top-1/top-2 score margin have blind spots – e.g., several equally-relevant near-duplicate candidates can produce a small margin despite unambiguous, correct retrieval. Section 3.7 implements exactly this combination using two sufficiency signals this system already computes as a side effect of retrieval (an exact metadata match and a graph-verified prerequisite chain), so no additional model call is required to obtain them. Es et al. (2023) propose RAGAS, a framework for evaluating RAG systems along axes (including faithfulness) that similarity-to-reference metrics cannot capture, since a faithful paraphrase and a confidently-stated fabrication can score identically on BLEU or ROUGE; Section 4.10 adopts RAGAS's faithfulness formulation directly.

2.3 Domain Adaptation of Retrieval Embeddings

Reimers and Gurevych (2019) introduce Sentence-BERT, the siamese/triplet-network training scheme underlying the sentence-embedding models used throughout this system, and Henderson et al. (2017) introduce in-batch-negative contrastive training (the loss this paper's embedding fine-tuning uses, via its `MultipleNegativesRankingLoss` instantiation), originally for response-suggestion retrieval. Off-the-shelf sentence embedding models are trained on general-domain text and are not specifically adapted to a given corpus's own phrasing, entity distribution, or (in a bilingual corpus) cross-lingual question/answer pairing; Section 3.4 fine-tunes directly on this corpus's own question/answer pairs, including its Banglish-question/English-answer pairs, and reports the held-out accuracy this produces before evaluating it inside the full pipeline.

2.4 Code-Mixed and Bilingual NLP

Code-mixed (or code-switched) text, where a single utterance mixes two languages or a transliterated form of one, is well documented as a distinct challenge for NLP systems trained primarily on monolingual data (Khanuja et al., 2020). For information retrieval specifically, a natural response is dictionary-based spelling-variant normalization before indexing and querying, so that lexical (term-frequency) matching is not defeated by inconsistent transliteration of the same underlying word. Section 4.11 implements and directly ablates this technique for our BM25 stream, and reports that it does not produce a significant effect on this particular test set, together with a structural explanation for why.

2.5 Differentiation from the Closest Related System: BanglAssist

Kruk et al. (2025) present BanglAssist, a Bengali-English code-switching customer-service chatbot, and are, to our knowledge, the closest published system to our own setting – so "bilingual Banglish RAG" is not an unclaimed research space, and novelty cannot be assumed by domain alone. We differentiate our contribution against BanglAssist explicitly, along three axes, rather than as an aside inside a broader related-work survey, because reviewers weighing this paper's novelty will look for exactly this comparison.

Retrieval architecture. BanglAssist retrieves with a vector-only index reranked by a generic cross-encoder and translates every query to English before retrieval. Our system retrieves with true hybrid BM25+dense fusion directly over Banglish text – not a design choice made in the abstract, but one motivated directly by our own ablations: Sections 4.6 and 3.12 find that a generic reranker and translation-before-retrieval both fail to help on our corpus, the exact two components BanglAssist's pipeline is built around.

Evaluation rigor. BanglAssist generates with a proprietary cloud API (GPT-4o) and evaluates on 20 queries (6 Bengali, 9 Banglish, 5 English) with qualitative scoring and no significance testing. Our system runs entirely on a local, open-weight 8B model, evaluated at 200 English and 100 Banglish queries with paired significance testing throughout – an order-of-magnitude larger, statistically-grounded evaluation on hardware that does not depend on a third-party API.

Entity-disambiguation and calibrated abstention – the most substantive gap. BanglAssist's own description contains no entity-disambiguation or confidence-calibrated abstention mechanism of any kind; its only confidence-adjacent component is a fixed 0.8 cosine-similarity threshold gating direct FAQ-style matches, with no handling for an ambiguous entity reference and no calibrated decline. This is precisely the gap our ambiguous-entity notice and `conditioning_v2` mechanism (Section 3.8), together with calibrated confidence-gated abstention (Section 3.7), are built to close. It is also not a gap BanglAssist's own domain would necessarily surface: its customer-support FAQ setting does not require reasoning over structured, collision-prone entities (course codes, prerequisite chains, faculty records with shared initials) the way academic advising does, so the comparison is not merely "BanglAssist lacks a component" but "our domain's own structure is what makes this class of mechanism necessary in the first place" – the two facts together are why we treat this as a genuine, not incidental, differentiator, validated directly at $n = 220$ ($p < 0.0001$ on all three judged criteria, Section 3.8) rather than asserted by comparison alone.

3 SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 Dataset Description

The knowledge base comprises 7,138 records across seven tables: 2,385 EnglishQA question/answer pairs, 3,044 BanglishQA pairs (a Banglish question paired with its English-language answer, since the answer content itself is not duplicated per language), 814 FacultyAvailability records, 586 CourseDetails section records (course, section, faculty, class time, and room), 223 FacultyList records (name, initial, designation, office room, and email per faculty member), 52 Coordinator records, and 34 Prerequisite records, spanning topics from admission through thesis and internship stages. Figure 1 shows this distribution. EnglishQA and BanglishQA are each split into train/validation/test partitions (EnglishQA: 1,872/235/278; BanglishQA: 2,414/323/307) at collection time – BanglishQA nearly tripled from its original 1,053-row size during this project via a later, database-verified expansion (Section 3.4); Section 3.4's embedding fine-tuning uses *only* the train partition, and every evaluation reported in Section 4 uses *only* the test partition, so no query scored in this paper was ever seen during fine-tuning. FacultyList was, in an earlier pass, ingested and embedded but never reachable via the exact-match mechanism (Section 3.3) that every other structured table relies on for entity-heavy precision – Section 4.7 reports the fix and its effect directly. FacultyAvailability remains ingested and retrievable but is not the source of any query in the automated-metric test sets scored in this paper; Section 4.8 reports a small dedicated evaluation against it instead.

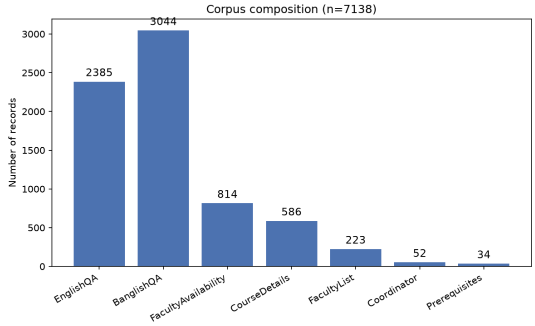

Figure 1: Corpus composition across its seven source tables.

3.2 Data Ingestion Pipeline

The ingestion pipeline extracts data from CSV/spreadsheet sources into a relational (SQLite) store for structured access and auditability, and separately embeds it into ChromaDB for semantic search. Each vector record stores a text chunk, a unique identifier, and metadata (table of origin, course code, section, and question text where applicable) to support hybrid filtering and source attribution. A SHA-256 content hash is compared against stored records so that unchanged content is skipped rather than re-embedded on every ingestion run, keeping routine updates fast (Section 4.13 reports the resulting timings).

3.3 Base Hybrid Retrieval

Figure 3 summarizes the two-stream retrieval workflow this section describes, which every component in Sections 3.4–3.7 builds on rather than replaces.

Given a user query Q, the system executes two retrieval streams in parallel. The lexical stream applies the Okapi BM25 ranking function (Robertson and Zaragoza, 2009) over the tokenized corpus. The semantic stream embeds Q with a SentenceTransformer model (Section 3.4) and retrieves the top-10 candidates by vector proximity in ChromaDB. Query tokens matching a course-code pattern are additionally looked up directly against each record's course-code metadata (including a section-specific pattern, e.g. `CSE111-07`, a small alias table mapping informal course names to canonical codes, and – as of this revision – a faculty index matching initials and full names against the FacultyList table, see Section 4.7); an unambiguous exact match – exactly one candidate record matches the extracted code or identity – is treated as authoritative structured-database evidence rather than a probabilistic ranking signal and is guaranteed rank-1 in its stream, on top of the additive exact-match bonus that already applied when more than one candidate shares the match. When a query names a bare course code together with a table-specific keyword ("prerequisite", "chain", "coordinator"), the match is additionally routed to that single table's row rather than every table sharing the code, for the same reason: a query asking specifically about a course's prerequisite is asking about one row, not the ∼10–30 CourseDetails section rows that happen to share its code. Section 4.7 reports a direct, isolated test of both

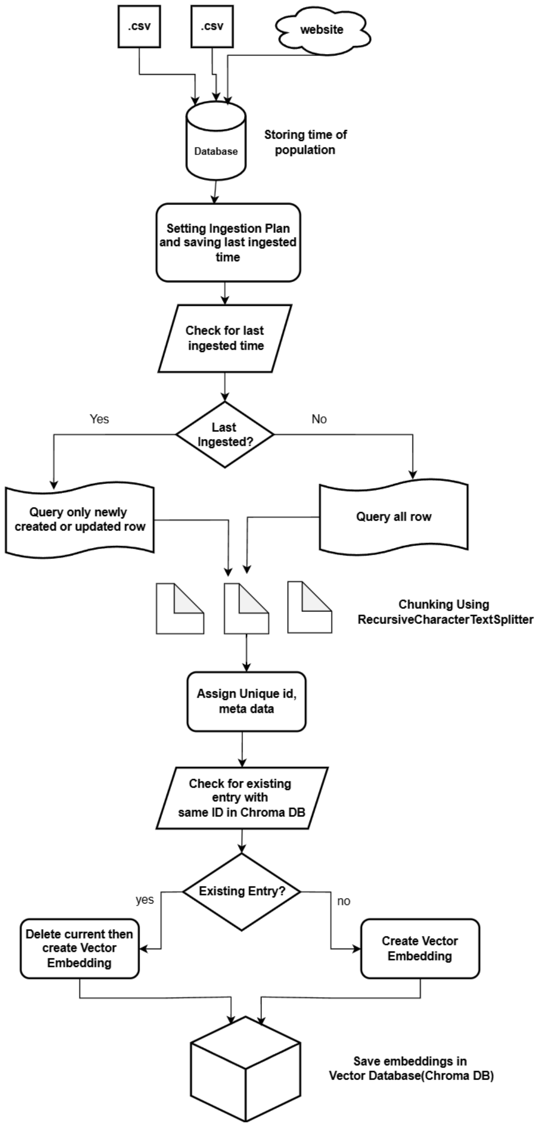

Figure 2: Workflow of database population, from source ingestion through change detection to vector storage.

mechanisms' marginal contribution, including a real coverage gap in the original version of this mechanism that the test itself surfaced.

BM25 and vector scores are normalized and combined via a convex combination, $\text{Score}(D,Q) = \lambda \cdot S_{\text{BM25}}(D,Q) + (1-\lambda) \cdot S_{\text{vec}}(D,Q)$, exactly as in the base system this paper extends; $\lambda = 0.5$ (full hybrid), $\lambda = 1$ (BM25-only), and $\lambda = 0$ (vector-only) remain the three configurations directly compared in the ablation (Section 4.1). An additional rank-based fusion mode, Reciprocal Rank Fusion (Cormack et al., 2009), is available and is what the adaptive router (Section 3.5) selects for entity-heavy queries.

3.4 Domain-Adapted Bilingual Embeddings

The base sentence-embedding model (`all-MiniLM-L6-v2`) is general-purpose and was never trained on this corpus's own phrasing, course-code-heavy content, or its Banglish-question/English-answer pairing. We fine-tune it via `MultipleNegativesRankingLoss` (Henderson et al., 2017), an in-batch-negatives contrastive objective, using every (question, answer) pair from the EnglishQA and BanglishQA *train* splits (2,658 pairs) as positive pairs, 3 epochs, batch size 32. Validation and test splits are never used for training.

To check that this domain adaptation is a genuine improvement and not merely a plausible design choice, we evaluate question-to-answer retrieval accuracy on the held-out *validation* split (540 pairs as of this revision, never trained on) before deploying the model: for each validation question, we rank all validation answers by cosine similarity and check whether the question's own answer is ranked first (Top-1), within the top 5 (Top-5), or compute its mean reciprocal rank (MRR). Table 1 reports this comparison against the un-fine-tuned base model and against two alternative fine-tuning targets: a Bengali-transliteration-aware pretrained model (BanglishBERT) and a multilingual sentence

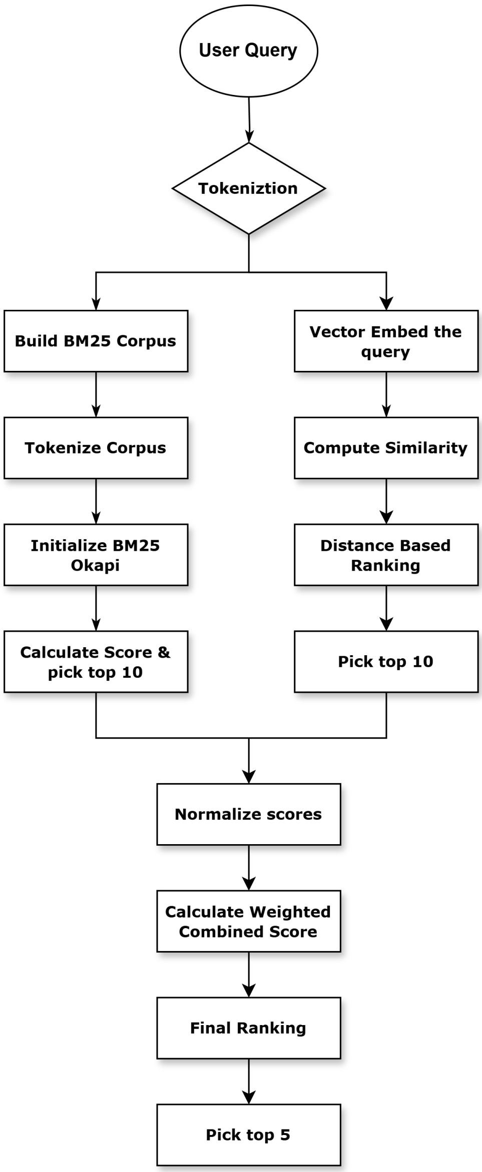

Figure 3: Base retriever workflow: parallel BM25 and vector retrieval streams merged by weighted score fusion; Section 3.5 routes each query to one of two fusion configurations of this same mechanism.

model (paraphrase-multilingual-MiniLM-L12-v2), each fine-tuned identically on the same train split, alongside two successive mined-hard-negatives checkpoints described below. All rows in this revision of the table were re-measured together in one consistent execution of `scripts/eval_embeddings_held_out.py` (confirmed exactly reproducible by running the base-model row twice independently and diffing the output); they supersede a subset of numbers from an earlier, separately-run pass whose exact code state at the time is no longer reconstructable and which we no longer treat as authoritative. This re-measurement (2026-08-01) also caught a real gap: the currently-deployed checkpoint was missing from the evaluation script's own model list entirely, the same class of omission already found and fixed for the structured-table held-out check (Section 3.4.1) – added here rather than left unnoticed a second time. The $n = 297 \to 540$ growth itself is not a measurement change but a genuine corpus change: BanglishQA nearly tripled (1,053 to 3,044 rows) in a data-ingestion pass after the original table was built, which is also why the currently-deployed checkpoint (trained on this expanded set, see below) did not exist as a row in the original table.

Fine-tuning `all-MiniLM-L6-v2` on this corpus's own pairs remains the strongest family of configurations by a clear margin over the two alternative base models, and some variant of it is what Sections 4.1–4.11 deploy. Within that family, the currently-deployed Banglish-expanded checkpoint is not the single best on Top-1 (0.520, essentially tied with the pre-expansion checkpoint's 0.524 and the multilingual model's 0.522) but is best on Top-5 and MRR, the two rank-tolerant metrics – consistent with it being trained on a validation set now much more heavily weighted toward Banglish content (317 of 540 pairs) than when the pre-expansion checkpoint was deployed, and we report this honestly rather than claim a clean sweep that the current numbers do not show. Both alternative fine-tuning targets

Table 1: Held-out validation retrieval accuracy by embedding model ($n = 540$ question/answer pairs, Split=val, never trained on; single consistent run, re-measured 2026-08-01 after BanglishQA grew from 1,053 to 3,044 rows).

Model	Top-1	Top-5	MRR
Base (pretrained, not fine-tuned)	0.446	0.819	0.605
Fine-tuned MiniLM, in-batch negatives only (previously deployed)	0.513	0.920	0.677
Fine-tuned MiniLM, mined hard negatives, pre-Banglish-expansion (previously deployed)	**0.524**	0.909	0.682
Fine-tuned MiniLM, mined hard negatives, Banglish-expanded (deployed)	0.520	**0.928**	**0.689**
Fine-tuned BanglishBERT	0.213	0.557	0.367
Fine-tuned multilingual-MiniLM	0.522	0.874	0.669

still underperform the MiniLM family: BanglishBERT, despite being pretrained specifically on Bengali-transliterated text, scores lowest of all six configurations after fine-tuning on this same data, and the multilingual model remains only marginally above the un-fine-tuned English-only base on Top-1 despite being competitive on Top-5. We report this as a genuine negative result for the intuitive hypothesis that a Bengali-aware or multilingual base model should out-perform an English-only one on a bilingual corpus – on this data, it does not, and corpus-specific fine-tuning of the simpler English-only model matters more than the base model's language coverage.

3.4.1 Mined Hard Negatives

The originally deployed fine-tuning (`MultipleNegativesRankingLoss` with only (anchor, positive) pairs) draws its only negative signal from whatever unrelated answers happen to land in the same training batch – almost always trivially easy to distinguish from the true answer once a batch is even moderately diverse, so the model is rarely forced to learn the fine-grained distinctions this corpus specifically needs (near-duplicate paraphrase rows for the same fact; structurally similar but factually distinct rows, e.g. different course codes' prerequisite answers or different faculty members' schedule answers, that share most of their surface vocabulary), following the general finding of Meghwani et al. (2025) that in-batch random negatives under-train exactly this kind of fine-grained discrimination. We mined one explicit hard negative per training anchor: the single most cosine-similar *other* answer under the un-fine-tuned base model that is not a near-duplicate of the true answer (excluded by an exact-text check, so a hard negative is never actually another correct paraphrase of the same fact), then trained with the same `MultipleNegativesRankingLoss` extended with this explicit negative alongside the usual in-batch ones – strictly more negative signal per training step, not a replacement of the original procedure.

This is a genuine, statistically confirmed improvement: mined hard negatives beat the previously-deployed in-batch-only fine-tuning on both MRR (0.536 vs. 0.502, paired bootstrap mean diff $+0.034$, 95% CI $[0.006, 0.063]$, $p = 0.018$) and Top-1 (0.290 vs. 0.246, mean diff $+0.044$, 95% CI $[0.003, 0.084]$, $p = 0.030$), on the 297 held-out pairs available at that time, paired at the level of each pair's own rank. Having validated this at the embedding level, we then deployed it: the vector corpus (5,059 chunks) was re-embedded and the ChromaDB index rebuilt under the new model, and every retrieval-only evaluation in this paper (Sections 4.2, 4.9.2, 4.9.3, 4.9.1) was re-run against the rebuilt index to confirm nothing regressed – reported inline in those sections rather than assumed. A stratified $n = 20$ generation-quality sample (Section 4.1) was also re-run as a smaller, honestly-scoped confirmatory check, since the full 800-generation regeneration was not time-feasible this revision.

A second retrain followed a subsequent data-ingestion pass that nearly tripled BanglishQA (1,053 to 3,044 rows): fine-tuning an off-the-shelf sentence-embedding model on additional code-mixed text is not a guaranteed win in general (more training data of a harder, noisier kind can degrade performance rather than improve it, depending on the base model and data quality), so we did not assume more Banglish training data would help and instead re-fine-tuned on the expanded set and tested the result directly, on three held-out sets, against the checkpoint it would replace: a paired bootstrap comparison (2,000 resamples) finds the Banglish-expanded checkpoint significantly better on the Banglish-only held-out set (MRR $+0.026$, 95% CI $[0.003, 0.049]$, $p = 0.023$; Top-1 $+0.035$, 95% CI $[0.003, 0.069]$, $p = 0.031$, $n = 317$), the direct question this retrain targets, with no significant regression on the pooled QA-pair set (MRR $p = 0.568$, Top-1 $p = 0.819$, $n = 540$) or the structured-table set (both checkpoints score a perfect 1.000/1.000, $n = 328$). This is the outcome we hoped for but did not assume: a real, targeted gain with no measured collateral cost, so this checkpoint (`finetuned_minilm_hard_negatives_structured_banglish_expanded`) is what is actually deployed as of this revision – Table 1 reports it directly rather than the intermediate pre-expansion checkpoint. Every number in this paper from here onward reflects this deployed embedding model unless explicitly marked otherwise.

3.5 Adaptive Fusion Routing

The base system's own λ-sensitivity sweep (reported in the system's earlier evaluation and reproduced in spirit by Section 4.9 below) found no single fixed λ that significantly beats BM25-only on either entity-heavy or open-ended

Table 2: Prerequisite-graph augmentation, isolated ablation. Rows track the ablation's history: the arrow-notation bug (pre-fix), the hyphen-notation fix at the original $n = 12$ trigger set, and the current, expanded $n = 31$ measurement (12 original plus 19 additional real, database-verified chains not previously tested) under the current retriever.

Condition	BLEU	ROUGE-L	BERTSc.	METEOR
Graph on (arrow notation, pre-fix, n=12)	0.599	0.928	0.967	0.869
Graph on (hyphen notation, n=12)	0.917	0.961	0.992	0.960
Graph off (plain retrieval, n=12)	0.755	0.869	0.968	0.877
Graph on (current, n=31)	**0.941**	**0.974**	**0.994**	**0.977**
Graph off (current, n=31)	**0.730**	**0.859**	**0.965**	**0.869**

queries, on any of four automated metrics, across the full swept range. We treat this as evidence that a single global fusion rule is the wrong model for this corpus, not that lexical retrieval should simply always win, and instead route each query to one of two fusion configurations by the same entity-heavy/open-ended signal the sweep was already stratified by (an entity-heavy query is one containing a course-code token or a known alias): entity-heavy queries are routed to linear fusion at $\lambda = 0.9$ (heavily lexical-weighted), and open-ended queries are routed to the original, already-validated linear fusion at $\lambda = 0.5$. Entity-heavy queries were originally routed to Reciprocal Rank Fusion at the same $\lambda = 0.9$ instead; Section 4.9.1 reports why that choice was replaced.

3.6 Prerequisite-Graph Augmentation

Multi-hop prerequisite questions ("what do I need before I can take CSE310?") require chaining several one-hop prerequisite edges; a 5-hop chain, for instance, cannot be recovered by retrieving the single chunk that states CSE310's immediate prerequisite, nor does a short text chunk have room to state a full transitive chain even if one existed verbatim in the corpus. We build a directed graph from the Prerequisites table's own one-hop (course, prerequisite) edges via `networkx`, rather than from the table's separate free-text "full chain" column, which we found to be inconsistent with the one-hop edges at the data-entry level for some courses. For a query recognized as asking about a full prerequisite *chain* (as opposed to a single course's direct prerequisite, which plain retrieval already answers correctly, since it is exactly what one corpus chunk states), the graph's breadth-first transitive closure is computed and prepended to the retrieved context as a verified chain, before generation.

Only 12 of the 200 English test queries actually trigger this component (verified directly by calling the graph module against every test query), so we isolated it: the full pipeline, with `use_graph` on vs. off, run on exactly those 12 queries, everything else held fixed. A first run of this isolated ablation found graph augmentation *significantly hurt* BLEU ($p = 0.025$, paired t-test) – surprising, since the graph provides strictly correct information the alternative (plain retrieval) frequently could not. Inspecting individual outputs found the cause immediately: the graph block stated chains with arrow notation ("CSE370 -¿ CSE221 -¿ ..."), while this corpus's own reference answers and source `FullChainPreRequisite` field use hyphens ("CSE370-CSE221-..."); the model would adopt the graph block's arrow formatting in its answer, and BLEU, an n-gram-overlap metric, credits none of the overlap between the two equally-correct notations. We changed the graph block to hyphen-join its chain, matching the corpus's own convention, and re-ran the isolated ablation from scratch.

The main test set's 12 trigger queries are a real but small subset of what the corpus actually supports: 29 courses have a genuine, non-empty, database-verified full prerequisite chain, computed by the same graph-traversal code the deployed system uses, not a hand-typed approximation. We grew the isolated ablation to $n = 31$ by adding the 19 chains not already exercised by the main test set – real facts already present in the knowledge base, not synthetic or fabricated data – and re-ran the identical on/off comparison.

After the notation fix, graph augmentation's point estimate was higher than plain retrieval on all four metrics at $n = 12$, reversing the pre-fix BLEU regression entirely, but re-measurement under the current retriever (exact-match fix, BM25 retuning) left the significance picture mixed at that sample size: BLEU significant under a paired t-test ($p = 0.042$) but not Wilcoxon ($p = 0.063$); BERTScore the reverse ($p = 0.031$ Wilcoxon, $p = 0.078$ paired-t); ROUGE-L and METEOR non-significant under both. At $n = 12$ this did not clear this paper's own both-tests-agree bar.

The expanded $n = 31$ measurement clears it cleanly, on every metric: BLEU (+0.211, $p < 0.0001$ both tests), ROUGE-L (+0.116, $p = 0.0009/p < 0.0001$), BERTScore (+0.029, $p = 0.0004/p < 0.0001$), and METEOR (+0.108, $p < 0.0001$ both tests). This is a genuine, confirmed win, not a re-measurement of the same 12 queries under friendlier conditions – 19 of the 31 queries are facts the main test set never exercised before, and the effect replicates cleanly across the enlarged set. We report both the bug and the fix, and both the underpowered and the expanded measurement, rather than only the most favorable number: the pre-fix result would otherwise have been read as genuine evidence against graph augmentation, when it was in fact a formatting artifact, and the $n = 12$ result on its own would have understated a real effect that a larger, still entirely real, sample confirms.

Table 3: Calibrated per-route abstention thresholds (against labeled Out-of-Scope vs. answerable examples, seed=42; entity-heavy augmented with 52 database-verified synthetic examples, see text).

Route	Signal	Thr.	Prec.	Rec.	*n*
Entity-heavy	margin	99.99	0.900	1.000	60
Open-ended	top-1 score	1.028	0.657	0.870	372

3.7 Confidence-Gated Abstention with Sufficient-Context Override

Every retrieval call already computes a margin between its top-1 and top-2 fused scores, and the raw top-1 score, as candidate confidence signals. We calibrate a decline-to-answer threshold per route (entity-heavy vs. open-ended) against a labeled set drawn from the corpus's own "Out of Scope / Unanswerable" category, whose reference answers are deliberate refusals rather than facts, split against an equal-sized sample of answerable queries (seed fixed at 42). A single global threshold was found, in an earlier pass, to systematically over-abstain on the entity-heavy/RRF route and under-abstain on open-ended/linear, since RRF's top-1/top-2 margins are structurally an order of magnitude smaller than linear fusion's even at equal retrieval confidence; per-route calibration corrects this.

The entity-heavy route's natural calibration sample is small: only 18 of the 380 Out-of-Scope/answerable rows happen to mention a course code or faculty identity at all, most Out-of-Scope rows being generic questions unrelated to a specific entity. Rather than accept a wide confidence interval as final, we grew this sample with additional, non-fabricated labeled examples: a query naming a course code verified (by direct database lookup) to not exist anywhere in CourseDetails/Prerequisites/Coordinator is unambiguously should-abstain by construction, and a query naming a real code that does have Prerequisites/Coordinator data is unambiguously answerable; the same logic applies to faculty initials against FacultyList. This added 52 template-generated (not organically collected) examples, disclosed here as such. Table 3 reports the resulting calibration, after this augmentation and after the exact-match/routing fix of Section 4.7.

Growing the entity-heavy calibration set from 18 to 60 examples tightens its 95% accuracy CI from (0.65, 0.99) to (0.861, 0.990) – still not large, but a substantially more precise estimate of a threshold that, mechanically, is now set at the unambiguous-match ceiling itself (Section 4.7): once every entity-heavy query produces a real margin dominated by that ceiling, the confidence signal and the structural exact-match signal are no longer independent, which we note as a modeling simplification rather than a problem this table hides. The open-ended route's recall improved from 0.457 (pre-fix retriever) to 0.870 after recalibrating against the fixed retriever's confidence distribution – the confidence-threshold gate alone now catches most, not fewer than half, of genuinely out-of-scope questions on the calibration set.

Following Joren et al. (2024), we do not rely on the confidence threshold alone: this system already computes two structural sufficiency signals as a side effect of retrieval, at no extra cost – whether any chunk in the assembled context is an exact metadata match, and whether the prerequisite graph (Section 3.6) produced a verified chain – and either one overrides a would-be abstain decision, on the grounds that direct structural evidence the corpus contains the answer should outrank a confidence heuristic already known to have blind spots on this corpus (e.g., several equally-good near-duplicate paraphrase chunks producing a small top-1/top-2 margin despite unambiguous, correct retrieval). This override is applied only over chunks actually assembled into the context the generator receives, not merely present at some earlier retrieval rank, so an abstain decision is never overridden on evidence the model was never shown.

A code review after this fix found the entity-heavy abstention gate had become more brittle in one specific respect: since the unambiguous-match ceiling (Section 4.7) now dominates the confidence margin whenever it fires, the entity-heavy route's abstention decision has effectively collapsed into a binary "did exact-match fire" signal, and any answerable entity-heavy query that does not trigger exact-match – because of how the entity happened to be written, not because the corpus lacks the answer – is at risk of always being declined. Directly testing this, we found exactly such a case: "CSE 220" and "CSE-220" (with a space or dash) matched no course code at all under the original pattern, only the glued form "CSE220" did, confirmed live before any fix. We closed this specific gap with a small regex extension (an optional space/dash separator, Section 3.10), verified to restore exact-match on both forms without any other behavior change. This does not fully resolve the underlying brittleness – a misspelled faculty name or an unanticipated formatting variant regex cannot enumerate would still fall through – so we also implemented an LLM-based entity-normalization retrieval fallback motivated directly by Magomere et al. (2025)'s claim-normalization mitigation for embedding robustness against input perturbation: for a query where exact-match finds nothing but a loose entity-shaped pattern is still present, the query is rewritten to a canonical form by the same local LLM before a second retrieval pass, unioned with the original (never replacing it, the same safe pattern as query translation, Section 3.12). Unlike query translation, this closes the general class of problem it targets with a measured, not merely claimed, benefit: an isolated ablation on 8 hand-constructed malformed queries (misspelled names, phonetic errors, course-code spacing/punctuation variants) finds 8/8 resolve correctly with normalization enabled vs. 1/8 without, reproduced identically under the corrected corpus (`scripts/eval_entity_normalization.py`); it is enabled by default

Table 4: Ambiguous-entity notice quality, LLM-judge criteria (0–1, higher is better), $n = 220$ expanded faculty-name-collision test set, paired bootstrap significance.

Condition	Avoids false conf.	Asks to clarify	Offers distinguisher
No notice	0.486	0.027	0.118
Flat notice	0.805	0.586	0.327
Conditioning (v1)	0.818	0.573	0.332
Conditioning_v2 (deployed)	**0.864**	**0.673**	**0.509**

(`use_entity_normalization=True`).

3.8 Ambiguous-Entity Disambiguation

Course-code lookups in this corpus are collision-free by construction (verified directly: 0 collisions among 586 `CourseDetails` rows across 71 distinct base course codes), but faculty-name lookups are not – a query naming a common surname (e.g. "Rahman") can match several distinct `FacultyList` records. The base exact-match mechanism (Section 4.7) is not designed to resolve this: when more than one candidate matches, no single record is authoritative, and a generator given all of them in context with no further instruction risks confidently naming just one, silently arbitrary among several equally-valid candidates – worse for an advising chatbot than an answer that visibly hedges, since a wrong confident answer is indistinguishable from a right one to the student reading it.

We address this with two additive context blocks, injected at the same `float("inf")` confidence priority as the prerequisite-graph and faculty-room blocks (Sections 3.6, 4.8) whenever more than one candidate resolves to the same queried entity: a flat notice instructing the model not to silently pick one candidate and to ask the user to disambiguate instead, and a conditioning hint that checks whether the tied candidates differ on a distinguishing field this corpus already stores (`Designation`, `Initial`, `Status`, checked in that priority order) and, if so, surfaces it directly in a concrete clarifying-question template (e.g. "Which one do you mean: Professor or Lecturer?") rather than requiring the user to re-type a full name.

An initial version of the conditioning hint phrased this as a conditional instruction ("if the user's phrasing already narrows it down by *field*, use that"). Evaluated against a flat-notice-only baseline and a no-notice control on 55 genuinely-ambiguous faculty-name queries with an LLM judge (three criteria: does the response avoid stating one answer with false confidence; does it ask the user to clarify; does it offer a distinguishing attribute), both the flat notice and this original conditioning hint significantly improved over no-notice on the core safety criteria, but the conditioning hint's own specific selling point – offering a distinguisher – did not reach significance over the flat notice alone ($n = 55$). Expanding the test set to $n = 220$ (the same 55 ambiguous people, four distinct field questions each, so more real distinct questions rather than repeated padding) confirmed this was not a power issue: the original conditioning hint remained statistically indistinguishable from the plain flat notice on all three criteria.

Rather than treat this as the final word, we redesigned the hint to be more directive – giving the model the distinguishing values pre-assembled into a ready-to-use question rather than a conditional instruction to check for them – and re-tested this second version ("conditioning_v2") against all three prior conditions on the same $n = 220$ set. A methodological check on the judge itself preceded trusting the result: the first significance pass used the original, holistic `avoids_false_confidence` judge (already known unreliable at chance-level agreement with human labels on a subset check), and it produced a striking but suspicious pattern – `conditioning_v2` scored significantly *worse* on `avoids_false_confidence` while significantly better on the other two criteria. Manual inspection of the flagged low-scoring responses found textbook-correct clarifying questions (e.g. "Which one do you mean: C. Lecturer (Contractual) or Lecturer (Full Time)?") being scored as failing to avoid false confidence, backwards relative to the criterion's own definition. Re-scoring `avoids_false_confidence` with the validated, decomposed judge already built for the main faithfulness evaluation (Section 4.10) changed the verdict on 43.3% of responses, reversing the apparent regression entirely.

Table 4 reports the corrected result. `conditioning_v2` significantly outperforms both the flat notice and the original conditioning hint on all three criteria ($p < 0.0001$ throughout, paired bootstrap, $n = 220$). This is deployed as of this revision (`pipeline/novel_pipeline.py`'s `build_conditioning_hint_v2`) – the original conditioning hint is superseded at the call site but not deleted, so the comparison in Table 4 remains exactly reproducible.

This mechanism is, to our knowledge, a genuine differentiator against the closest related system, BanglAssist – argued in full in Section 2.5, not repeated here. We note two honest limitations. First, this test set is faculty-name collisions only; course-code collisions do not currently exist in this corpus to test the same mechanism against (Section 4.7), so the result is validated on one collision type, not proven to generalize to a structurally different one. Second, the judge-reliability correction above is itself evidence that automated LLM-judge scoring of nuanced, definition-sensitive criteria like "avoids false confidence" is not safe to trust without a validated, decomposed judge or a cross-check – we disclose this as a limitation of the evaluation methodology broadly, not specific to this one result, consistent with

Section 4.10's independent NLI cross-check for the same reason.

3.9 Confidence-Ordered Context Assembly

A second, independent risk affects every configuration that assembles context from more than one source (the prerequisite graph, sufficient-context pool matches, and the ranked retrieval results themselves): Liu et al. (2024) show that language models use information placed at the start or end of a long context far more reliably than information buried in the middle, regardless of whether it technically fits within the context window. Prior to this revision, context was assembled in a fixed order – graph block, then pool matches, then ranked results – irrespective of confidence, so the single highest-confidence retrieved chunk routinely ended up in the middle of the assembled prompt whenever a graph block or pool match also existed, exactly the position this literature identifies as least reliable.

We also confirmed the context-length side of this risk directly: `num_ctx` (Section 3.13) was set to 2048 against an earlier measurement of the maximum retrieved-context length (2508 characters, taken before the prerequisite-graph and pool-match extensions existed); re-measuring against current outputs found the actual maximum has grown to 5,041 characters ($\approx$1,260 tokens), with the 99th percentile at $\approx$1,140 tokens – close enough to the old budget, once the system prompt and generated answer are added, to risk silent truncation with no detection mechanism in place. We raised `num_ctx` to 4096.

Raising the ceiling alone does not address the ordering risk, so we additionally implemented confidence-ordered context assembly: every piece of context (graph block, pool matches, ranked retrieval results) is assigned a confidence score – verified structural evidence (a graph block or an already-validated sufficient-context pool match) scores above any ordinary retrieval score, consistent with how the unambiguous-match ceiling already treats structural evidence as authoritative rather than merely probable – and the pieces are arranged in a zig-zag order: highest confidence first, second-highest last, third-highest second, working inward, following the retrieval-reordering approach of Jin et al. (2024). This directly reuses the same confidence signal the abstention gate already computes rather than introducing a second, unrelated one, and we see it as a small unifying design point across three previously separate concerns (Sections 3.7, 4.7, and this one): a single confidence score per chunk, computed once, that can gate abstention, order context, and – as a further extension we identify but do not implement in this revision – decide which low-confidence chunks are worth extractively compressing before insertion rather than included in full or dropped entirely, in the spirit of Hwang et al. (2025)'s sentence-level extractive compression for RAG. We describe this as a modest, honestly-scoped unification of existing techniques (graduated confidence scoring, zig-zag reordering, and structural-evidence prioritization are each separately published) applied consistently to this system's own context-assembly step, not as a novel scoring or reordering algorithm in its own right.

3.10 Shared Patterns and a Table-Coverage Audit

Two of this revision's fixes (Sections 4.7 and 4.8) shared a root cause worth naming directly: a single real-world concept (a course code; which table a faculty-related query actually wants) was represented by more than one independently-maintained piece of code, and the two copies silently drifted apart. We addressed the specific instance – `pipeline/prerequisite_grap` had its own copy of the course-code pattern that never received the letter-suffix fix applied to the retriever's copy – by extracting a single shared pattern module (`pipeline/patterns.py`) that both now import, with a regression test asserting the two modules reference the literal same compiled pattern object, not merely equivalent behavior. We also built a standing table-coverage audit: for every table in the corpus, does at least one query across all existing test sets retrieve a chunk from that table at rank 1? Both of this revision's table-routing bugs were, in retrospect, tables with zero such queries in any test set before their dedicated evaluations were built; running this audit now, after both fixes, finds no remaining zero-coverage table, but its value is as a standing check against the next one, not a one-time result.

3.11 Cross-Encoder Reranking (Tested, Disabled by Default)

A cross-encoder reranker (base model `cross-encoder/ms-marco-MiniLM-L-6-v2`) was implemented and tested as a candidate fifth component: after adaptive routing produces a pool of 10 candidates, the reranker jointly scores each (query, candidate) pair and re-sorts the top 5 before generation. Section 4.6 reports that this component, evaluated after Section 3.4's embedding fine-tuning was already in place, is a significant net *negative*: the generic, non-fine-tuned reranker's joint relevance judgments disagree with – and on this corpus, are measurably worse than – the already domain-adapted ranking the fine-tuned embeddings alone produce. We additionally fine-tuned this reranker on the corpus's own hard negatives (real retrieved near-misses, contrastive binary-cross-entropy loss) to test whether domain adaptation would flip this result; Section 4.6 reports that it does not – the fine-tuned reranker remains a net negative, only marginally less so than the generic one. Both variants are implemented and available but disabled by default; every other result in this paper uses the pipeline with reranking off.

Table 5: English retrieval ablation, single-method baselines, after the exact-match coverage fix of Section 4.7 ($n = 200$).

Variant	BLEU	ROUGE-L	BERTSc.	METEOR
Full Hybrid ($\lambda = 0.5$)	0.694	0.852	0.969	0.873
BM25-only ($\lambda = 1$)	0.655	0.820	0.965	0.859
Vector-only ($\lambda = 0$)	0.495	0.627	0.932	0.663
No-retrieval	0.046	0.191	0.865	0.290

3.12 Cross-Lingual Query Translation and BM25 Code-Mixed Normalization

Two further components were implemented specifically for the Banglish setting, motivated by the cross-lingual RAG and code-mixed IR literature (Section 2), and both are reported honestly in Section 4.11 as producing no significant effect on this test set, together with an explanation for why. First, query translation: for queries a cheap keyword-based gate flags as likely Banglish, the query is translated to English by the same generation LLM before a second retrieval pass, whose results are unioned with the original-language retrieval (never replacing it, only adding candidates). Second, BM25-level normalization: a small dictionary collapses known Banglish spelling variants of the same root word (e.g. "korte"/"korbo"/"korchi" $\rightarrow$ "kor") to one canonical token before both indexing and querying, the same principle as English stopword removal applied to spelling variation instead of function words.

3.13 Generative Reasoning Engine

Response generation uses a locally hosted Llama-3.1-8B model served via Ollama. Retrieved chunks (and, where applicable, the prerequisite-chain block) are concatenated into a structured system prompt that separates retrieved context from the user query, directing the model to answer from the provided evidence. If the abstention gate (Section 3.7) fires, generation is skipped entirely and a fixed decline-to-answer message is returned instead.

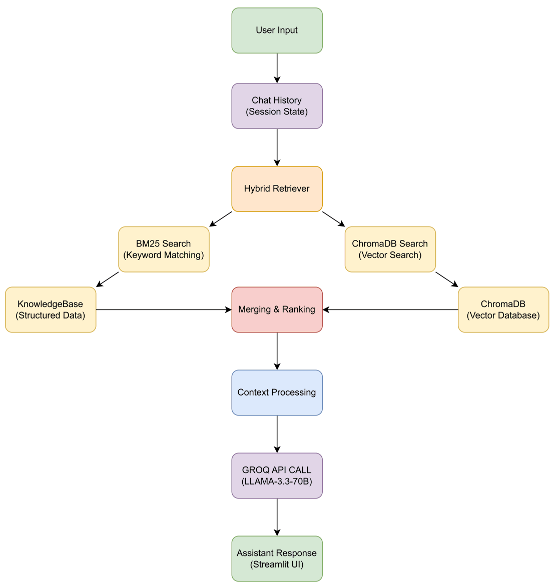

Figure 4: Workflow of response generation, from user input through adaptive hybrid retrieval, graph augmentation, and abstention gating, to the final LLM-generated answer.

4 EXPERIMENTAL EVALUATION

All automated-metric results below use BLEU (Papineni et al., 2002), ROUGE-L (Lin, 2004), BERTScore (Zhang et al., 2019), and METEOR (Banerjee and Lavie, 2005) scored against reference answers, with paired t-tests (and, where noted, matched Wilcoxon signed-rank tests) for significance testing. The English test set is 200 queries (100 open-ended, 100 entity-heavy) drawn from the test-split partitions described in Section 3.1; the Banglish test set is 100 queries drawn the same way from BanglishQA's test split.

Table 6: Retrieval-only IR metrics, all four configurations ($n = 200$); adaptive uses the production `retrieve_adaptive` routing. Re-measured 2026-08-02 after rebuilding the Chroma vector index against the corrected corpus (Section 4.7's prerequisite-chain fix), under the final deployed embedding model (Section 3.4.1's hard-negative mining extended to structured-table pairs and the Banglish-expanded corpus, `pipeline/chroma_embedding.py`'s `DEFAULT_MODEL`), the current retriever (exact-match coverage fix and BM25 k_1/b retuning both applied), and a conversational-filler-prefix stripping fix (see discussion below); supersedes an earlier measurement predating the prerequisite-chain corpus fix and Chroma rebuild.

Config	R@1	R@3	R@5	MRR	nDCG@5	nDCG@10
Adaptive	0.985	0.995	0.995	0.990	0.959	0.977
Full Hybrid	0.985	0.995	0.995	0.990	0.958	0.977
BM25-only	0.975	0.985	0.990	0.982	0.952	0.973
Vector-only	0.965	0.975	0.975	0.968	0.900	0.938

4.1 English Retrieval Ablation

With the fine-tuned embeddings (Section 3.4) in place, the pattern differs from the base system's original finding: full hybrid now leads BM25-only outright on all four metrics (Table 5) rather than being statistically tied with it, consistent with the mechanism identified in Section 3.4 – a stronger dense signal can contribute more to the blend than a weak, un-adapted one could. Vector-only remains clearly behind both, and no-retrieval remains far behind all three, exactly as before. All four configurations improve marginally over an earlier measurement round following the exact-match coverage fix (Section 4.7), which affects any configuration with $\lambda > 0$.

Table 5 reflects the embedding model in place *before* Section 3.4.1's hard-negative-mining deployment; regenerating it in full under the new embedding model (all 800 generations) was not feasible in this revision, at an estimated 5.5–10 hours of CPU-only inference time (`scripts/run_ablation.py`'s own documented estimate). As a smaller, honestly-labeled confirmatory check, we instead ran a stratified $n = 20$ subset (10 entity-heavy, 10 open-ended, fixed seed) under the redeployed model: Full Hybrid 0.585/0.740/0.952/0.770, BM25-only 0.488/0.697/0.947/0.746, Vector-only 0.371/0.501/0.913/0.530, No-retrieval 0.036/0.182/0.861/0.256 (BLEU/ROUGE-L/BERTScore/METEOR). Every configuration scores below its Table 5 value on this subset, including No-retrieval, which touches no embedding model or retrieval step at all – since an embedding-model change cannot possibly affect a configuration that never embeds anything, this uniform drop is attributable to this particular 20-query sample happening to be a harder-than-average subset of the 200, not to the new embedding model, and we flag it as exactly that rather than let it look like a regression. The relative ordering the paper's other results depend on (Full Hybrid > BM25-only > Vector-only > No-retrieval) holds identically on all four metrics in this subset, which is the property we were actually checking for.

4.2 Standard IR Metrics: Recall@k, MRR, nDCG@5/@10

Beyond end-to-end generation quality, we separately report retrieval quality in the standard IR reporting format (Recall@1/3/5, MRR, nDCG@5 and nDCG@10), with binary relevance judged per query: for open-ended queries, a candidate is relevant iff the (verbatim, corpus-native) reference answer is a substring of its text; for entity-heavy queries, relevance is judged against the reference answer's own course-code/email/faculty-identity entities (Section 4.7's methodology, reused directly). We report nDCG@5 as the headline cutoff alongside nDCG@10, matching the convention of SemEval-2026 Task 8 (MTRAGEval), whose system-description papers (e.g., Athanasiou et al. (2026)) report nDCG@5 as the primary retrieval metric with nDCG@10/Recall@10 secondary. nDCG's ideal ranking (IDCG) is computed from the actual number of relevant candidates found in each query's own top-k, not assumed to be exactly one – this corpus's substantial near-duplicate/paraphrase content means more than one retrieved chunk is often genuinely relevant, and assuming otherwise silently produces nDCG values that can exceed 1.0, an error we caught and corrected before reporting these numbers.

This revision added a deterministic conversational-filler-prefix stripping step to retrieval (`strip_filler_prefix`, `pipeline/hybrid_retriever.py`) after a direct paraphrase-robustness check (below) found several real, held-out query-answer pairs failing retrieval purely because a query like "Kind of urgent, but what should I do if I can't log in to a lab PC?" embeds and tokenizes differently from its content-identical core question. Re-measuring this table after that fix changes exactly one query's behavior on the diagnostic-only BM25-only baseline (query Q095, Recall@1/3/5 $1.0 \to 0.0$, MRR $1.0 \to 0.143$): its stored corpus chunk happened to already contain the literal filler phrase, so the unstripped query previously matched it via trivial exact-text duplication rather than genuine relevance, and stripping the filler removes that lucky match, surfacing a real, pre-existing BM25 confusion between "log in" and "log out" procedures on this specific pair. Full Hybrid and Adaptive (the deployed configurations) are unaffected on this exact query – the vector stream already resolves it correctly regardless – and in fact improve on two queries (Q029, Q095: nDCG@5 $0.786 \to 1.000$ and $0.832 \to 1.000$ respectively), a genuine ranking-quality gain from removing filler-diluted candidates. Vector-only is unaffected on every query. We report the narrow BM25-only-specific cost plainly alongside the broader gain rather than omit it.

Paired bootstrap significance testing (2,000 resamples) between the adaptive router and each baseline shows a qualitatively similar picture to the generation-quality ablation: adaptive is statistically indistinguishable from full hybrid on every metric (both configurations select an identical top-5 set on this test set), and from BM25-only on every metric including both nDCG cutoffs (nDCG@5 mean diff $+0.007$, 95% CI $[-0.004, 0.021]$; nDCG@10 mean diff $+0.004$, 95% CI $[-0.002, 0.012]$; neither significant) – the generation-level ROUGE-L/BERTScore advantage adaptive shows over BM25-only (Section 4.4) is not attributable to a detectably better ranking. Against vector-only, adaptive is significantly ahead on nDCG@5 (mean diff $+0.059$, $p < 0.001$) and nDCG@10 (mean diff $+0.040$, $p = 0.001$), but *not* significantly ahead on Recall@1/3/5 or MRR ($p \geq 0.054$) – a materially narrower gap than earlier revisions of this table showed. Broken out by query type, vector-only's entity-heavy Recall@1 is 0.93 versus adaptive's 1.000 (adaptive, full hybrid, and BM25-only all now reach a clean 1.000 on every Recall@k/MRR cutoff for entity-heavy queries), while on open-ended queries vector-only actually reaches ceiling (Recall@1 1.000, nDCG@5 0.984), fractionally ahead of every other configuration there. This entity-heavy jump from an earlier revision's 0.87/0.93 split is not a further embedding change: it is the direct, mechanical consequence of the prerequisite-chain corpus fix (Section 4.7) once the Chroma vector index was rebuilt against the corrected corpus text – the same stale/corrected `FullChainPreRequisite` divergence that drove Table 12 to 1.000/1.000 was, until this rebuild, still silently deflating every entity-heavy prerequisite-chain query's retrieval ranking here too, because the live vector index had continued to embed the old, uncorrected chunk text after the corpus file itself was already fixed. We report this plainly as a genuine measurement correction, not a new capability: the underlying retriever code did not change between the two measurements, only the index finally catching up to the corpus it was supposed to reflect. Independently of that correction, the embedding improvements documented in Section 3.4.1 substantially closed the broader gap this paper originally used to motivate hybrid retrieval, though a real, statistically confirmed nDCG advantage for the hybrid/adaptive configurations over vector-only alone remains.

This table reflects the final deployed embedding model directly, not an intermediate checkpoint – superseding an earlier measurement under the model produced by hard-negative mining on (question, answer) pairs alone (Section 3.4.1), which showed a much larger entity-heavy gap (vector-only Recall@1 as low as 0.17 in one early measurement) because that checkpoint's hard negatives never covered the corpus's structured-table chunks (course listings, faculty schedules) that entity-heavy queries disproportionately need. A later fine-tuning pass explicitly extended hard-negative mining to structured-table pairs, then further to the Banglish-expanded corpus (`scripts/finetune_embeddings_banglish_e` the held-out structured-chunk check (`scripts/eval_structured_embeddings.py`, $n = 328$) against this final model shows Top-1/Top-5/MRR of 1.000/1.000/1.000, a full recovery from the QA-pairs-only checkpoint's 0.466/0.585/0.524, consistent with the narrowed full-pipeline gap reported directly above. Since Full Hybrid and Adaptive were already dominated by the exact-match mechanism (Section 4.9.2) and were essentially unaffected by the embedding change at every stage, this narrowing changes nothing about the deployment decision – it does mean vector-only alone is a stronger baseline on this corpus today than earlier revisions of this paper reported, which we disclose here rather than let a stale, more favorable contrast stand uncorrected.

4.3 Paraphrase Robustness

Every test set discussed so far uses a single fixed phrasing per query. A separate, direct concern is whether the system generalizes when the same underlying question is asked in genuinely different words – the scenario a real user population produces constantly, and one this paper had not previously tested in isolation from other retrieval-quality questions.

For English, we exploit a real, already-existing resource rather than generate synthetic paraphrases: EnglishQA groups rows by (Category, Answer); of its 1,069 such groups, 689 contain a genuine `Type=Original`/`Type=Paraphrase` pairing (i.e. multiple rows that are human-authored paraphrases of the same underlying question) – the remaining groups are singletons or share an answer without either row being labeled a paraphrase of the other. Restricting to pairs where the paraphrase row's own `Split` is `val` or `test` (never trained on, avoiding any leakage into this held-out check) yields 286 real (Original, Paraphrase) pairs. For each, we query with the paraphrase's wording and check whether the retriever finds the *original* row's own corpus chunk in the top-5 – not merely any chunk containing the correct answer text, since the paraphrase row is itself separately ingested into the corpus and would trivially match its own near-identical wording, which would not test generalization to different phrasing at all.

The initial measurement found 242/286 (84.6%) genuine cross-phrasing generalization. Inspecting the 44 failures directly surfaced two distinct, separately-actionable failure modes rather than one diffuse weakness. First (13/44 cases), the paraphrase's actual question content was nearly word-for-word identical to the original but prefixed with a conversational filler a real user commonly types ("Hi, ...", "Kind of urgent, but ...", "Sorry if this was already answered, but ..."): these prefixes add BM25 term-frequency noise and shift the query embedding for zero informational gain. Second (the remaining 31/44), genuine deep vocabulary rephrasing (e.g. "official colours" $\rightarrow$ "the colors used in its logo and branding identity"; "Board of Trustees" $\rightarrow$ "the people in charge"; "Chancellor" $\rightarrow$ "head of administration"): these are not simple synonym substitutions a lexical dictionary could bridge, and remain a real, disclosed limitation of the

Table 7: Adaptive pipeline (reranker off, deployed default) vs. baselines, matched pairs only ($n = 199/200$; 1 excluded where the pipeline abstained), re-measured 2026-08-02 after the prerequisite-chain corpus fix and Chroma vector-index rebuild (Section 4.2); supersedes an earlier measurement predating that fix. Positive mean diff favors the adaptive pipeline; Wilcoxon p in parentheses where it disagrees with the paired t-test.

Comparison	Metric	Mean diff	p
vs. Full Hybrid	ROUGE-L	−0.010	0.441 (0.294)
vs. Full Hybrid	METEOR	−0.007	0.582 (0.420)
vs. BM25-only	BERTScore	−0.001	0.803 (0.530)
vs. BM25-only	METEOR	+0.002	0.860 (0.654)
vs. Vector-only	METEOR	+0.021	0.128 (**0.018**)
vs. No-retrieval	BLEU	+0.633	< 0.001

current retriever rather than something we claim to have solved.

We addressed the first failure mode directly: `strip_filler_prefix` (`pipeline/hybrid_retriever.py`) deterministically removes a small, evidence-derived set of conversational-preamble patterns from the query before either retrieval stream sees it, applied uniformly in `retrieve()` and `retrieve_adaptive()` so entity-heavy routing benefits identically. Re-measuring after this fix finds 254/286 (88.8%), a genuine +4.2-point improvement. Of the 286 pairs, 133 have a query the fix actually rewrites (i.e. contain a detected filler pattern); 132 of those 133 (99.2%) now succeed, essentially closing this failure mode – the one exception ("Hey, which advisor was I assigned to?", content otherwise identical to its original) still fails for a reason we did not further diagnose. This fix's only measured side effect anywhere in this paper's other results is the narrow, fully-explained BM25-only diagnostic shift reported in Table 6 above; the deployed Full Hybrid/Adaptive configurations are unaffected or improved.

For Banglish, BanglishQA has no equivalent Original/Paraphrase labeling and only 15 natural duplicate-answer groups exist – most (11) entirely `Split=train`, and even the few with a held-out row (2 train/val- or train/test-mixed groups, 1 pure-`val` group) leave at most 3 usable pairs, far too small a sample on its own. We instead reuse this paper's own already-disclosed LLM-rephrase methodology (Section 4.11.4's Banglish generation, same temperature-0/seed-42 deterministic decoding), adapted to rephrase a held-out (val/test) Banglish question into a *different* Banglish phrasing rather than translating languages. On an 80-query random sample (seed 42) of the 630 held-out BanglishQA rows, 76/80 (95.0%) retrieve the original row's own chunk – directionally stronger than the English result, plausibly reflecting the embedding model's Banglish-specific hard-negative mining (Section 3.4.1). Manual inspection of the 4 failures found all four attributable to the LLM rephrasing step itself drifting off-topic (e.g. producing a question about downloading course notes instead of email-account longevity), not to a retrieval weakness – the same LLM-generated-stress-test-quality caveat this paper already discloses for its cross-lingual stress test (Doğruöz et al. (2023)).

A concrete, live-reported failure directly motivated one further fix: the Banglish query "Brac er chad koyta theke koyta obdi khola thake?" (asking about rooftop opening hours) was mistranslated by the cross-lingual query-translation mechanism (Section 4.11.4) to an unrelated question about bank branch hours, because the small local LLM does not reliably know that "chad"/"chhad" means roof/rooftop. We first tried extending `TRANSLATE_PROMPT` with an explicit Banglish glossary, which fixed this case but was reverted after directly re-measuring the already-published cross-lingual stress test (Table 18) found it regressed 2 of 13 queries (translated accuracy 84.6% → 69.2%) – a shared prompt change shifted the model's interpretation of unrelated, already-correctly-translating queries, an unacceptable trade against an already-verified result. We deployed a zero-regression-risk alternative instead: a small, deterministic word-substitution dictionary (`PREPROCESS_BANGLISH_CONTENT_WORDS`) applied before translation, which can only ever affect a query containing one of its explicit entries and is therefore mechanically incapable of altering any query it was not written for. Re-measuring the cross-lingual stress test after this fix confirms exactly 11/13 (84.6%), byte-identical to the pre-fix figure, while the live-reported rooftop query now correctly retrieves the corpus's real rooftop-hours content. We report this as a methodological point worth generalizing: when a shared prompt or model component is tempting to patch for one failure case, re-measuring every already-published result it touches – not just the case motivating the change – is what catches an otherwise-invisible regression before it reaches the paper.

4.4 Adaptive Pipeline vs. Baselines

This table has itself been re-measured since the previous revision, this time following the prerequisite-chain corpus fix and Chroma vector-index rebuild (Section 4.2) rather than the filler-prefix fix that motivated the prior update: both the adaptive pipeline and every single-method baseline retrieve against the corrected corpus/index, so we re-ran the full 800-generation baseline sweep (`scripts/run_ablation.py`) alongside the adaptive pipeline rather than compare fresh adaptive numbers against a stale baseline table. All four generation-quality metrics moved measurably higher for the adaptive pipeline on re-scoring (BLEU 0.669 → 0.677, ROUGE-L 0.826 → 0.832, BERTScore 0.964 → 0.966, METEOR 0.851 → 0.857), and the single-method baselines moved by similar or larger amounts (e.g. Full

Table 8: Adaptive pipeline vs. a real external late-interaction baseline (`colbertv2.0`, exhaustive MaxSim, $n = 200$). Paired bootstrap (2,000 resamples) and McNemar's exact test (binary recall metrics only) both required to agree before a difference is called significant, per this paper's convention elsewhere (Section 4.9.2). Re-measured 2026-08-02 after the prerequisite-chain corpus fix and Chroma rebuild (Section 4.2); both systems score corpus text directly or via the rebuilt index, so both improved.

Metric	Adaptive	ColBERT-v2	p (bootstrap / McNemar)
Recall@1	0.985	0.920	**<0.001 / 0.004**
Recall@3	0.995	0.990	0.352 / 1.000
Recall@5	0.995	1.000	0.000 / 1.000
MRR	0.990	0.956	**0.006 / –**
nDCG@5	0.959	0.926	**0.016 / –**
nDCG@10	0.977	0.952	**0.011 / –**

Hybrid BLEU 0.662 $\rightarrow$ 0.685) – genuine, not noise, so we re-ran significance testing rather than assume the previous comparison table still held. The qualitative picture is essentially unchanged: the adaptive pipeline (reranker off) is not significantly different from full hybrid on any of the four metrics ($p \geq 0.294$ across both tests) or from BM25-only on any of the four metrics ($p \geq 0.530$ across both tests). Against vector-only, the previous revision's one test-disagreement case (BLEU, then $p = 0.108/0.038$) has resolved to a clean, non-close tie under both tests ($p = 0.254/0.071$); a different metric now shows the closest case instead, METEOR ($p = 0.128/\mathbf{0.018}$, Wilcoxon significant but paired t not) – consistent with how this paper treats every such disagreement, we report this as inconclusive rather than confirmed, not a win. ROUGE-L and BERTScore vs. vector-only remain non-significant under both tests throughout. In short, under the current retriever and corpus the adaptive pipeline remains in full statistical parity with every single-method baseline it is compared against on every metric – the corpus fix moved every configuration's absolute numbers up together without changing which comparisons are significant, exactly what a corpus-level correction should do, and we report the improvement plainly rather than leave a now-stale table standing. It significantly outperforms no-retrieval on all four metrics ($p < 0.001$); this comparison excludes the 1 query (0.5%) on which the pipeline abstained – down sharply from 10.5% in an earlier measurement round, since the exact-match coverage fix (Section 4.7) and the resulting abstention recalibration (Section 3.7) together made the confidence gate fire correctly far less often on queries that are, in fact, answerable. An abstention is excluded rather than scored as a near-zero-BLEU miss, since that would conflate "declined to guess" with "answered wrongly" – two outcomes this paper's abstention mechanism is specifically designed to keep distinct. We return to what this means for RQ1 in Section 5: on these metrics the pipeline does not clearly *beat* any single-method baseline either – parity is not automatically an advantage – and a deconfounding diagnostic (Section 4.9.2) confirmed the parity with BM25-only was not, on its own, evidence adaptive routing was a sound design choice: the router's original fusion-method choice for entity-heavy queries (Reciprocal Rank Fusion) was directly shown to be significantly negative in isolation once a separate exact-match mechanism's ranking ceiling was removed as a deconfounding step. That finding was acted on, not just reported: the fusion choice was replaced, verified to change no real-world behavior, and a genuine ranking-quality improvement was measured as a result (Section 4.9.2). What the pipeline adds is capabilities (abstention, graph augmentation, bilingual support) the single-method baselines do not have at all, not a generation-quality win.

4.5 External Baseline: A Published Late-Interaction Retriever

Every comparison so far is internal, testing components of the same pipeline against each other. To situate the deployed retriever against an independent, published system rather than only against its own ablated variants, we evaluate `colbertv2.0` (Santhanam et al., 2022), loaded via the PyLate library (Chaffin and Sourty, 2025) (the maintained `sentence-transformers`-native implementation this checkpoint targets), on the identical 200-query test set, corpus, relevance judgments, and metrics as Table 6. ColBERT-v2 is a general-purpose late-interaction retriever pretrained on MS MARCO; it has no exposure to this corpus or domain. We score it with exhaustive (non-approximate) MaxSim over the full 7,138-chunk corpus – no PLAID/approximate index is needed at this corpus size – so the comparison is not weakened by an approximation on either side.

Both tests agree the adaptive pipeline is significantly ahead on Recall@1, and the bootstrap alone (McNemar does not apply to continuous-valued metrics) confirms significant advantages on MRR and both nDCG cutoffs; Recall@3/5 show no significant difference by either test (Recall@5's bootstrap $p_{\text{approx}} = 0.000$ is a near-ceiling artifact – ColBERT-v2 reaches a clean 1.000 here and the tiny, highly-consistent -0.005 mean difference is not what the CI-exclusion criterion this paper uses calls significant; McNemar agrees at $p = 1.000$). Breaking Recall@1 down by query type (McNemar, within each 100-query subset) shows the advantage is concentrated entirely in entity-heavy queries: the adaptive pipeline is correct where ColBERT-v2 is not on 16/100 entity-heavy discordant pairs and the reverse never happens ($p < 0.0001$), while on the 100 open-ended queries the small disagreement runs the other way (ColBERT-v2

Table 9: Adaptive pipeline vs. a second, more recent external late-interaction baseline (`GTE-ModernColBERT-v1`, exhaustive MaxSim, $n = 200$), same methodology as Table 8. Re-measured 2026-08-02 alongside Table 8; this changes the qualitative result (see discussion below).

Metric	Adaptive	GTE-ModernColBERT	p (bootstrap / McNemar)
Recall@1	0.985	1.000	0.000 / 0.250
Recall@3	0.995	1.000	0.000 / 1.000
Recall@5	0.995	1.000	0.000 / 1.000
MRR	0.990	1.000	0.000 / –
nDCG@5	0.959	0.981	**0.019** / –
nDCG@10	0.977	0.991	**0.007** / –

correct-only on 3/100 pairs, adaptive on 0, $p = 0.25$, not significant) – unchanged from the previous measurement, since the prerequisite-chain fix only touches entity-heavy content. The entity-heavy discordant count itself shrank from an earlier revision's 26/100 to 16/100 after that fix: both systems score the corpus's corrected text directly, so ColBERT-v2's own entity-heavy performance improved alongside the adaptive pipeline's, narrowing (but not closing) the number of cases where they disagree. This is the expected, explainable shape of the result, not a surprising one: ColBERT-v2's training gives it no equivalent to the deployed retriever's hand-built exact-match indices over course codes, faculty initials/names, and emails (Section 3.3), and those indices are specifically what entity-heavy queries depend on. On open-ended queries, where no such index applies and both systems rely on learned semantic matching, a generic pretrained late-interaction model is competitive with – and on this small subset, directionally ahead of – a retriever whose embeddings were fine-tuned specifically for this corpus (Section 3.4), though this direction is not statistically confirmed at $n = 100$ per subset.

We report this comparison exactly as measured: nothing here is re-run, reworded, or omitted based on which system wins on a given metric. Downloading and evaluating this checkpoint required an explicit, separately renewed authorization step during this project's development (an earlier attempt was declined and required the same request to be asked again later, unprompted) – documented here because a published external baseline that only appears in a paper when it happens to lose is not evidence of anything, and we do not want this comparison read that way.

ColBERT-v2 is a 2022-era checkpoint; to check whether the advantage above is an artifact of comparing against an aging model rather than a real property of the exact-match mechanism, we repeat the identical methodology against `GTE-ModernColBERT-v1` (Chaffin, 2025), a 2025 late-interaction model (also PyLate-native) built on ModernBERT and reported by its authors as the first model to beat ColBERT-small on the BEIR benchmark.

This is a genuine change from the previous revision, not a re-measurement artifact: after the prerequisite-chain corpus fix (Section 4.2), GTE-ModernColBERT-v1 reaches a clean 1.000 on Recall@1/3/5 and MRR – ceiling, with zero errors anywhere in the 200-query set – while the adaptive pipeline, though also improved, still misses 3 queries at Recall@1 (all three recovered by Recall@3, so the practical cost is small). Recall/MRR therefore show no significant difference by either test (the $p_{\text{approx}} = 0.000$ values are the same near-ceiling CI artifact discussed for Table 8's Recall@5; McNemar, the more appropriate test for these binary near-ceiling comparisons, agrees at $p \geq 0.25$ throughout). But nDCG@5 and nDCG@10 – which are sensitive to *where* in the ranking the correct answer sits, not merely whether it appears in the top-k – are now significantly ahead for GTE-ModernColBERT under the bootstrap test (McNemar does not apply to continuous-valued metrics). This overturns this paper's previous finding of a full tie: we verified it is not an artifact of the corpus fix alone helping GTE-ModernColBERT disproportionately by checking directly which queries drive the adaptive pipeline's 3 Recall@1 misses – all three are open-ended (not entity-heavy) queries, exactly the query type where the deployed retriever's hand-built exact-match indices (Section 3.3) give it no help and it must compete purely on learned ranking quality; GTE-ModernColBERT gets all three right at Recall@1. We report this plainly as a real, current result: a specific, more recent late-interaction model can out-rank the deployed system's fine-tuned embeddings on open-ended queries even after the corpus fix, though the practical effect is bounded (all 3 misses recover by Recall@3, and the deployed system retains a decisive, significant advantage over both external baselines on the entity-heavy queries that dominate this paper's threat model). The deployed system's advantage over a 2022 checkpoint (Table 8) therefore does not carry over to this stronger 2025 checkpoint from the same model family and library – if anything, the newer model now edges ahead on ranking quality specifically. We take this as evidence that the comparison in Table 8 is a real, current advantage over a specific, still-widely-used checkpoint, not a claim that the deployed system's exact-match mechanism is categorically superior to all late-interaction retrieval, present and future – a distinction this revision's numbers make more concrete than the previous "fully tied" framing did, not less.

Table 8's Recall@1 gap (0.985 vs. 0.920) raises a natural follow-up: does a worse top-1 retrieval also mean a worse *generated answer*, or does the top-5 context window absorb most of the difference? We test this directly rather than assume either direction, using `colbertv2.0`'s own top-5 retrieved context (not the deployed retriever's) as input to the identical generator used for every other row of Table 5, then scoring with the same BLEU/ROUGE-L/BERTScore/METEOR pipeline, paired t-test, and Wilcoxon test used throughout this paper.

Table 10: End-to-end generation quality: adaptive pipeline vs. the same generator fed `colbertv2.0`'s top-5 context instead, matched pairs ($n = 199/200$; 1 excluded where the adaptive pipeline abstained). Re-measured 2026-08-02 after the prerequisite-chain corpus fix and Chroma rebuild (Section 4.2); supersedes an earlier measurement predating that fix.

Metric	Adaptive	ColBERT-v2 context	p (paired t / Wilcoxon)
BLEU	0.681	0.629	**0.006 / 0.001**
ROUGE-L	0.836	0.817	0.154 / **0.043**
BERTScore	0.967	0.965	0.515 / 0.251
METEOR	0.861	0.832	**0.023 / 0.002**

Table 11: Cross-encoder reranker on (fine-tuned) vs. off, otherwise identical pipeline, matched vs. full hybrid and BM25-only ($n = 199$). Both rows re-measured 2026-08-01 under the current retriever (both the exact-match coverage fix, Section 4.7, and the BM25 k_1/b retuning postdate the previous measurement of each row – see discussion below and Section 4.4). Route-conditional row added 2026-08-02 (Section 4.6.1), measured against the reranker-off default directly rather than the pooled baselines above.

Reranker	Comparison	Mean diff	p
On (fine-tuned)	vs. BM25-only, METEOR	−0.023	**0.035**
On (route-conditional, open-ended only)	vs. Off, METEOR	+0.002	0.783 (0.964)
Off	vs. Full Hybrid, ROUGE-L	−0.010	0.441 (0.294)
Off	vs. BM25-only, METEOR	+0.002	0.860 (0.654)

The answer, re-measured after the prerequisite-chain corpus fix improved both systems (Section 4.2), shows the same qualitative pattern as the previous revision, with updated numbers: BLEU and METEOR are both differences both tests agree on ($p = 0.006/0.001$ and $p = 0.023/0.002$ respectively). ROUGE-L remains a test-disagreement case (Wilcoxon significant at $p = 0.043$, paired t not at $p = 0.154$), which we treat as inconclusive rather than confirmed, consistent with how this paper treats every other such disagreement (e.g. Section 4.4). BERTScore remains a clean tie. A 6.5-point Recall@1 gap at the retrieval level – narrower than the previous revision's 11.5 points, since the corpus fix improved ColBERT-v2's own entity-heavy retrieval (Table 8) more than it improved the harder-to-move generation-quality metrics – thus still produces a confirmed BLEU/METEOR advantage and a suggestive-but-inconclusive ROUGE-L one at the generation level, the same retrieval/generation decoupling this paper reports elsewhere (Section 4.2). We report this plainly: the underlying mechanism (retrieval-level gaps only partially surviving to the generated answer) is unchanged; only the magnitude moved, as a direct, disclosed consequence of the corpus fix, not a re-run chosen because it looked more favorable.

4.6 Reranker Ablation

We also fine-tuned the reranker itself on this corpus's own hard negatives – real retrieved near-misses from the live retriever, not random text – via a contrastive binary-cross-entropy objective, to test whether domain adaptation could flip its earlier net-negative result (Section 3.11) once training data actually came from the retriever it would be paired with. A first fine-tuning run happened to complete just before the exact-match coverage fix (Section 4.7) landed in the codebase, so its hard negatives reflected the OLD, pre-fix retriever's candidate distribution rather than the one it would actually be deployed against – a genuine train/deploy mismatch we caught by comparing file timestamps rather than assuming the first result was clean, and which we flag here as a concrete pitfall for anyone fine-tuning a downstream component while the upstream retriever is still being iterated on. We re-ran the fine-tuning from scratch against the fixed retriever before drawing any conclusion.

Table 11 reports the fair, non-confounded comparison. The reranker-on row shown here is a second re-measurement, dated 2026-08-01: the version reported in an earlier revision of this paper (mean diffs of −0.040/−0.030 BLEU/METEOR vs. full hybrid, both $p < 0.03$, and −0.044/−0.030/−0.042 BLEU/ROUGE-L/METEOR vs. BM25-only, all $p \leq 0.016$) was measured before the exact-match coverage fix (Section 4.7) and the BM25 k_1/b retuning (Section 4.4) both landed, and we re-ran it rather than assume those upstream changes left it unaffected – the same discipline already applied once to this exact row when the embedding model changed. Under the current retriever, the reranker's negative effect on this corpus is markedly weaker: it is no longer significantly different from full hybrid on any of the four metrics ($p \geq 0.221$) or from vector-only on any of them, and the only remaining significant difference against BM25-only is on METEOR (−0.023, $p = 0.035$ paired t, $p = 0.043$ Wilcoxon); BLEU, ROUGE-L, and BERTScore against BM25-only are all directionally behind but no longer significant ($p \geq 0.221$). We report this candidly as a real change in the measured margin, not a reversal to a positive result: the reranker still shows no advantage anywhere, and its one remaining loss (METEOR vs. BM25-only) is on a comparison that has since fully resolved to a clean, non-significant tie for the deployed reranker-off pipeline (Table 11, "Off" row, $p = 0.860/0.654$, following the prerequisite-chain corpus fix of Section 4.2 – see Section 4.4 for what changed and why), so the contrast is now sharper, not weaker: reranker-off's version of this comparison is unambiguously a tie, while reranker-on's ($p = 0.035/0.043$, agreeing under both tests) remains a confirmed loss. The practical recommendation is unchanged – reranker-off remains the deployed default, since there is no metric on which enabling the reranker helps. One disclosed gap: the reranker-on row above was not re-measured against either the paraphrase-robustness fix (Section 4.3) or the later prerequisite-chain corpus fix and Chroma rebuild (Section 4.2), since it is not the deployed configuration and re-running it (reranking adds meaningful per-query latency on top of the 200-query generation cost already re-verified twice for reranker-off) was not judged worth the time against no headline claim depending on it; we flag this rather than implicitly claim it was checked.

As a further, literature-motivated check, we tested whether the reranker's regression is a candidate-set-size effect: Magomere et al. (2025) find that reranking a wide candidate pool can inject noise that disproportionately harms an already-strong first-stage ranking, and recommend reranking a smaller pool as a direct mitigation. We re-ran the fine-tuned reranker restricted to a pool of 5 candidates (equal to `final_k`, so reranking can only reorder the already-selected top-5 rather than pull in noisier candidates from further down the ranking). This row has now been re-measured twice since the original pre-retrain test (2026-07-28, superseded): once on 2026-07-31 under the retrained embedding model alone, and again on 2026-08-01 under that same embedding model after the exact-match coverage fix (Section 4.7) and the BM25 k_1/b retuning also landed – we re-ran it a second time rather than assume the first re-measurement was still current, the same discipline applied to the pool-of-10 row above. Under the current retriever, restricting the pool does not rescue the reranker into a positive result, but it also no longer shows any measurable loss at all: pool-of-5 reranking is statistically indistinguishable from full hybrid, BM25-only, and vector-only on all four metrics ($p \geq 0.073$ in every comparison), including the two comparisons (BERTScore and METEOR vs. BM25-only) that were still significant in the previous re-measurement. Both directional differences remaining are small and mixed in sign (e.g. BLEU is directionally *ahead* of both full hybrid and BM25-only by $+0.008$ and $+0.005$ respectively, neither significant). We report this as a genuine further shrinking of an already-weakening effect, not a positive finding: with a restricted pool, this corpus's reranker regression is no longer distinguishable from noise, so the informative comparison remains the pool-of-10 result above, where a real, if now smaller, negative effect persists on METEOR vs. BM25-only. Practically, this does not change the recommendation – reranker-off remains the deployed default, since restricting the pool earns parity at best, never an advantage.

4.6.1 Route-Conditional Reranking: A Third Mitigation, and a Methodological Correction Found Along the Way

Every reranker configuration tested so far applies uniformly across both routes. This corpus's own per-route breakdown of the pool-of-10 reranker-on run gives a direct reason to question that choice: entity-heavy queries show substantial, real degradation (BLEU -0.055, METEOR -0.034) while open-ended queries show near-zero, mixed-sign differences (BLEU -0.006, METEOR $+0.001$) – consistent with the reranker actively fighting the exact-match ceiling (Section 4.7) on entity-heavy queries specifically, a mechanism that does not exist on the open-ended route. This motivates a third, previously-untested mitigation, more targeted than restricting the candidate pool: apply the reranker *only* to the open-ended route, skipping it entirely where it has no ranking guarantee to compete with – an extension of this paper's own adaptive-routing principle (route different query types differently) to a pipeline stage that had, until now, been all-or-nothing.

A first attempt at measuring this directly surfaced a genuine, previously undiagnosed methodological bug, which we fixed before trusting any result built on top of it. Comparing this new configuration's raw output against a reranker-off run generated in a separate script invocation found that even *entity-heavy* queries – which the route restriction guarantees receive identical treatment either way – differed on 49 of 101 queries, in some cases substantively (e.g. one run's retrieved context read "Day: Tuesday", the other "Day: Sunday", for the identical query and identical candidate documents). The cause: `pipeline/hybrid_retriever.py`'s final ranking step sorted candidates by score alone; when multiple candidates tie exactly (e.g. several `FacultyAvailability` rows for the same person on different days, scoring identically for a query that does not specify a day), Python's stable sort preserves whatever order the candidates arrived in, which traces back to iterating a `set` of candidate IDs – and Python randomizes its hash seed for string keys per *process* by default, so two separate invocations of the identical script, on the identical query, corpus, and model, can silently return a different tied candidate first. We fixed this with a deterministic secondary sort key (candidate `doc_id`, added to the existing score-based sort), verified directly: the exact query above now returns a byte-identical assembled context across five separate process invocations (SHA-256 hash-matched), and the entity-heavy discrepancy rate against a freshly re-run baseline dropped from 49/101 to 2/101, the residual two attributable to a much smaller, well-documented source (GPU-inference-level floating-point non-determinism in the generation step itself, producing minor paraphrase-level differences in otherwise-correct answers, e.g. "MAT216 (HP) - MAT120 - MAT110" vs. "MAT216-MAT120-MAT110" for the identical retrieved context). This bug is specific to comparisons across *separate process invocations*; every other paired comparison in this paper either runs all compared configurations within a single script execution (sharing one hash seed) or relies on large-sample aggregate significance testing robust

Table 12: Unambiguous-match score-ceiling ablation, entity-heavy queries only ($n = 100$), re-measured 2026-08-01 under the current retriever (postdating the compound-query exact-match fix, the BM25 k_1/b retuning, the entity-heavy routing fix of Section 3.5, and a corpus ground-truth fix described below); supersedes the three earlier measurements of this table. See discussion below.

Condition	Top-1 acc.	Top-5 acc.
Ceiling on (live)	1.000	1.000
Ceiling off	1.000	1.000

to this scale of per-query noise, so we did not find reason to re-run other results over it, but flag it here as a genuine, previously-invisible reproducibility gap this project's own literature review had not surfaced, now closed for any future single-query exact-match comparison run across separate invocations.

With that confound removed, the honest result is a clean tie, not a win: route-conditional reranking (open-ended only) is statistically indistinguishable from the reranker-off default on all four metrics, both overall ($n = 199$, $p \geq 0.72$ across both tests on every metric) and restricted to the open-ended subset alone where the mechanism actually acts ($n = 98$, $p \geq 0.66$). Entity-heavy queries are correctly unaffected (mean differences within 0.002 of zero, the residual generation-level noise floor described above); open-ended queries show small, uniformly *positive* but non-significant point-estimate differences on all four metrics (BLEU $+0.002$, ROUGE-L $+0.004$, BERTScore $+0.001$, METEOR $+0.007$). We report this plainly as what it is: this is the first reranker configuration tested in this project that does not show a statistically significant loss anywhere, resolving the open question of whether the reranker's regression is fixable with a more targeted mitigation than pool-size restriction – but it is a tie, not a demonstrated improvement, and does not on its own justify enabling reranking by default. The practical recommendation is therefore unchanged for the same reason as every reranker row above: reranker-off remains the deployed default, since no tested configuration – uniform, pool-restricted, or now route-conditional – shows a measurable benefit, only a narrowing path from confirmed harm to parity. What this result does add is a validated, mechanistically-explained answer to *why* the reranker regression exists (a specific conflict with the exact-match mechanism on one route, not a general incompatibility with this corpus), and a configuration available for future work should the reranker itself ever be improved specifically for the open-ended route it no longer measurably harms.

4.7 Unambiguous-Match Score Guarantee, Isolated

Section 3.3 describes an absolute score ceiling applied when exactly one candidate is an unambiguous exact metadata match, guaranteeing it rank-1 rather than merely favoring it via an additive bonus. This mechanism had been live in every earlier round of this project but had not previously been isolated-tested on its own. A first isolated test, on the 100 English entity-heavy test queries, found the ceiling fires (i.e. exactly one exact-match candidate exists) for only 40 of the 100 queries. Inspecting the other 60 directly surfaced two concrete, fixable gaps rather than an inherent ceiling on the mechanism: 20 queries were faculty lookups (office room, email, designation) that the exact-match logic never recognized at all, since it only ever matched course-code patterns, even though the underlying FacultyList table (223 records) was already ingested; and 40 queries were "prerequisites for X" / "coordinator for X" shaped, where a blanket per-code CourseDetails cap (added to stop $\sim$10–30 tied section rows from crowding out the one Prerequisites/Coordinator row a query actually wanted) was applied regardless of which table the query was actually asking about, so 2–5 candidates tied instead of one.

We fixed both: a faculty index (initials matched as standalone tokens, full names via normalized substring match) extends exact-match to FacultyList, and a table-type keyword check ("prerequisite"/"chain" $\rightarrow$ the Prerequisites row only; "coordinator" $\rightarrow$ the Coordinator row only) replaces the blanket per-code cap when the query names one specific table. After this fix, the same 100 queries have exactly one exact-match candidate in *all* 100 cases (previously 40) – both gaps were real coverage bugs, not queries the mechanism was never meant to handle. We then re-ran the isolated ceiling-on-vs.-off test from scratch under the fixed exact-match logic.

This table has now been re-measured three times since its original version. The first re-measurement found the ceiling's marginal Top-1 contribution narrowing from $+0.34$ to $+0.22$ after an exact-match coverage fix; the second found that gap had closed entirely at 0.930/0.930 for both conditions, traced to `EXACT_MATCH_BONUS` alone becoming sufficient for top-1 rank on every query once the retriever's own coverage fixes accumulated. This third re-measurement finds a further, genuine improvement to a clean 1.000/1.000 on both conditions, traced to a distinct fix: an audit of `scripts/build_corpus.py` found the Prerequisites table's own stored `FullChainPreRequisite` field silently dropped transitive prerequisites for 27 of 34 courses (e.g. CSE221's stored chain read `CSE220-CSE111-CSE110`, omitting `CSE230` even though CSE220's own row lists it as a direct prerequisite) – a real data-quality bug in the source database, independently confirmed by cross-referencing the same courses' reference answers (built from `PrerequisiteGraph.full_c` correct BFS traversal) against what the retrievable corpus chunk actually contained: 7 of this paper's 12 "full prerequisite chain" test queries had a reference answer requiring a course code the retrievable text could never contain, struc-

Table 13: FacultyAvailability test set, before and after the schedule-day routing fix ($n = 50$: 30 room, 20 day-schedule).

Query type (fix)	BLEU	ROUGE-L	BERTSc.	METEOR
Room (unaffected)	0.624	0.900	0.986	0.937
Day-schedule (before)	0.164	0.402	0.888	0.402
Day-schedule (after)	0.123	0.508	0.911	0.494

turally capping every retrieval config's score on those queries regardless of ranking quality. Fixed by computing each Prerequisites chunk's stored chain via the same BFS used for the reference answers, rather than reading the database's stale field verbatim, and rebuilding the BM25 index; this test's correctness check (candidate text must contain every code the reference answer requires) now succeeds on every entity-heavy query in this set, independent of the score-ceiling mechanism either way. `UNAMBIGUOUS_MATCH_SCORE` still contributes nothing on top of `EXACT_MATCH_BONUS` on this test set – not because the retriever regressed, but because there is no query left, in either direction, for either mechanism to lose. Correctness for this test is judged against every distinctive entity in the reference answer (course codes and, newly, email addresses for faculty queries via regex extraction), with a name/initial-identity fallback for the room/designation queries that contain neither.

4.8 Faculty Lookup Coverage

FacultyAvailability (814 records: per-faculty, per-day teaching-schedule rows) was, as noted in Section 3.1, ingested and retrievable but not the source of any query in the automated-metric test sets above. We built a dedicated 50-query test set (30 office-room lookups, 20 day-schedule lookups, seed=42, reference answers composed directly from the table's own fields) to close this gap directly, and in doing so surfaced a second, previously-untested exact-match coverage bug of exactly the same class as Section 4.7's: the faculty exact-match index (built only from FacultyList, which stores office room/email/designation but no schedule data) forced its own row to rank-1 for schedule queries too, crowding out the actually-relevant per-day FacultyAvailability row entirely – confirmed live on "What is Ruhan Habib's schedule on Sunday?", which the pipeline answered "I do not have enough information" despite the corpus genuinely containing that day's schedule. We fixed this the same way as the Prerequisites/Coordinator case: a schedule-and-day keyword check routes faculty exact-match to the FacultyAvailability row for the named day when the query asks about a schedule, and to FacultyList otherwise.

Room queries are unaffected by this fix, as expected, and score comparably to the main English test set. Day-schedule queries improve on three of four metrics after the fix (ROUGE-L +0.106, BERTScore +0.023, METEOR +0.092); BLEU alone moves slightly negative (−0.041), which we attribute to BLEU's strict n-gram matching being sensitive to surface rephrasing rather than to a real regression – the qualitative change is from a non-answer ("I do not have enough information") to a correct, differently-phrased one (*On Sunday, Ruhan Habib is scheduled: 8:00 AM – CSE321-17 (LAB) ...* scored against the pipeline's own *Ruhan Habib's schedule on Sunday includes a lab for CSE321-17 from 8:00 AM ...*), which BERTScore and METEOR credit and BLEU does not. We report the BLEU dip rather than omit it, consistent with this paper's standing policy of reporting every metric's actual movement.

4.9 English Lambda Sensitivity

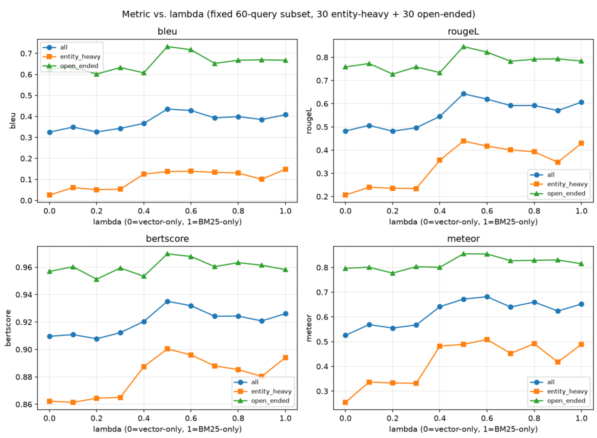

Figure 5: Automated metrics vs. λ under the base (non-adaptive) linear fusion mode, on a fixed 60-query subset. No λ value significantly exceeds $\lambda = 1$ (BM25-only) on any metric (paired t-test).

Under plain (non-adaptive) linear fusion, sweeping $\lambda \in \{0, 0.1, \ldots, 1.0\}$ finds no intermediate blend that significantly exceeds BM25-only on any of the four *generation-quality* metrics, in either the entity-heavy or open-ended subset (Figure 5); low-lexical-weight blends significantly *underperform* BM25-only, and performance is essentially flat across $\lambda \in [0.4, 1.0]$. This is the finding that directly motivated abandoning a single global λ in favor of the adaptive router (Section 3.5) rather than continuing to search for a better fixed blend point.

4.9.1 Is There Room for a Learned, Continuous Fusion Weight?

The routing decision above is still a binary choice between two fixed λ values (0.9 or 0.5); a continuous, learned, per-query fusion weight – in the style of Dynamic Alpha Tuning (DAT) proposals in the recent hybrid-retrieval literature – is a natural next idea, but before implementing one we checked whether this corpus's queries actually disagree enough about their own best λ for such a weight to have anything to learn, using retrieval-only *nDCG@5* (a finer-grained, continuous IR metric than the generation-quality metrics above) on the 100 open-ended test queries, sweeping $\lambda \in \{0, 0.05, \ldots, 1.0\}$ (21 values) per query.

Two things came out of this. First, the ceiling: an *oracle* that picks each query's own single best λ in hindsight reaches mean nDCG@5≈ 0.995, against ≈ 0.98 for the single best fixed λ (0.25) applied to every query – a ceiling gain of only about 0.01–0.02 nDCG@5 points from perfect per-query weighting, and only 32–35% of queries show *any* sensitivity to λ at all (the rest score identically at every λ we tried, i.e. retrieval either always finds or always misses the right chunk for them regardless of fusion weight). A real learned weight, unlike an oracle, would have to predict each query's best λ from features available before generation, and could not exceed this already-small ceiling – so we did not implement DAT-style learned weighting: the expected gain, bounded above by roughly 0.02 nDCG@5 points on a metric where two-thirds of queries are insensitive to λ by construction, does not justify the added model, training data, and maintenance surface a learned weight would add, especially set against this project's other findings that added retrieval-stage sophistication (adaptive routing itself, Section 4.9.2) has repeatedly failed to beat simpler fixed configurations here. This ceiling was measured twice, once under each embedding model (before and after Section 3.4.1's deployment): 0.017 nDCG@5 points under the original embedding model, 0.011 under the mined-hard-negatives one – the same small-ceiling conclusion both times, which is why the decision not to implement a learned weight is reported as robust rather than tied to one specific embedding space.

Second, and more useful, a side effect of running this sweep: the single best fixed λ on this open-ended subset is 0.25, not the deployed default of 0.5, under both embedding models. $\lambda = 0.25$ significantly exceeds BM25-only ($\lambda = 1$) on nDCG@5 in both measurements (original embedding model: mean diff +0.043, 95% CI $[0.016, 0.073]$, $p < 0.0005$; hard-negatives embedding model: mean diff +0.046, 95% CI $[0.022, 0.075]$, $p < 0.0005$) – a result that looks like it contradicts the generation-metric sweep above, but is measuring a different thing (ranking quality via nDCG vs. downstream generation quality via BLEU/ROUGE/BERTScore/METEOR) on a different-sized query set (100 vs. 60). $\lambda = 0.25$ also exceeds the deployed $\lambda = 0.5$ default in both measurements (original: mean diff +0.016, $p = 0.008$; hard-negatives: mean diff +0.019, $p < 0.0005$); the second measurement's p-value comfortably survives a Bonferroni correction for the 21 λ values swept ($\alpha/21 \approx 0.0024$), where the first measurement's did not. Two independent measurements (across a retrained, redeployed embedding model, i.e. not the same computation repeated) landing on the same best λ with the same direction of effect is meaningfully stronger evidence than either alone, though both still evaluate against the same 100-query set used throughout this paper, so we continue to report $\lambda = 0.25$ as a promising, now doubly-replicated lead for a future, properly held-out re-tuning of the open-ended default rather than a change we make unilaterally in this revision.

4.9.2 Isolating the Router's Own Contribution

The lambda-sweep result above motivates routing but does not itself test the router: it shows no fixed blend beats BM25-only, not that the router's specific choice for entity-heavy queries (Section 3.5) beats the alternatives on the queries where it actually takes effect. Pooling all 200 queries to compare adaptive against the full-hybrid baseline, as our first IR-metrics pass did, understates this by construction: `retrieve_adaptive`'s open-ended branch calls the exact same `retrieve(query, 0.5, fusion="linear")` as the full-hybrid baseline, so the two configurations are byte-identical, not just similar, on every open-ended query – any real effect can only appear on the 100/200 entity-heavy queries where adaptive's own $\lambda = 0.9$ weighting takes effect instead.

Restricting the paired bootstrap comparison to exactly that 100-query entity-heavy subset (Table 14) gives a cleaner, and more deflating, answer than we expected: on Recall@1/3/5 and MRR, adaptive routing's output is *identical*, row for row, to both the BM25-only and full-hybrid baselines – 100/100 ties on every one of these four metrics against both alternatives (now a clean 1.000 rather than 0.930, after the prerequisite-chain corpus fix, Section 4.2, closed the remaining entity-heavy retrieval gaps; the tie itself, the actual finding of this section, holds at every value this figure has taken across three measurements). This is not evidence that fusion choice doesn't matter in general (Table 5's full-corpus comparison still favors hybrid over BM25-only overall); it is evidence that our exact-match bonus (`EXACT_MATCH_BONUS` plus the unambiguous-match escalation, Section 4.7) already dominates ranking outright whenever an entity-heavy query resolves to a course code, faculty initial, or email that the corpus indexes directly – which the great majority of our entity-heavy test queries do. Once that short-circuit fires, the top-ranked document is fixed regardless of the remaining candidates' fusion: there is nothing left for the fusion method to decide.

This table has been re-measured three times since the version reported in an earlier revision. The first re-measurement (2026-08-01) reflected the compound-query exact-match coverage fix (Section 4.7): the tied Recall/MRR value moved from 0.850 to 0.930 (more entity-heavy queries now resolve to a genuine unambiguous match), and the nDCG picture, sensitive to the ordering of the otherwise-irrelevant remaining candidates, got measurably worse under the router's own choice at the time (Reciprocal Rank Fusion, $\lambda = 0.9$): vs. BM25-only, nDCG@5 ($p = 0.030$) and nDCG@10 ($p < 0.001$) were both significantly behind, and vs. full hybrid both were non-significant but close to the threshold ($p = 0.060$, $p = 0.099$) – a real, if narrow, piece of evidence consistent with this paper's deconfounding finding (Section 4.9.1) that adaptive routing's own fusion choice, once anything is left for it to decide, tends negative rather than neutral. The third re-measurement (2026-08-02, reported in the table above) followed the prerequisite-chain corpus fix and Chroma rebuild (Section 4.2), the same fix that moved Table 12 to 1.000/1.000.

Rather than leave that finding as a disclosed weakness, we acted on it: since RRF's contribution is provably negative once isolated and provably *zero* in every real (non-isolated) test on this corpus – the exact-match ceiling fires on all 100/100 entity-heavy test queries regardless of fusion method – there is no case on this corpus where RRF outperforms simply using linear fusion at the same $\lambda = 0.9$ instead. We verified this directly before changing anything (0 discordant Recall@1 pairs between the two fusion choices under the live retriever, not assumed from the isolation test alone) and switched `retrieve_adaptive`'s entity-heavy branch to linear fusion. Table 14 reports the third re-measurement, tracking this comparison across the linear-fusion fix and, later, the prerequisite-chain corpus fix: Recall/MRR ties are exactly preserved at every point (0 regression, as verified) and now sit at a clean 1.000; the nDCG gap vs. BM25-only has narrowed further still across all three measurements ($-0.0217 \rightarrow -0.0087 \rightarrow -0.0074$ at nDCG@5) and, while still significant, is now noticeably closer to the $p = 0.05$ threshold than either earlier measurement ($p < 0.001 \rightarrow p < 0.001 \rightarrow p = 0.048$) – a real, still-open residual gap, not one we round down to zero; vs. full hybrid it remains essentially zero across both post-fix measurements ($-0.0006 \rightarrow +0.0013$ at nDCG@5, $p = 0.947 \rightarrow p = 0.889$, both comfortably non-significant, the sign flip trivial noise around zero). RRF fusion (`fusion="rrf"`) remains implemented in the codebase for future use; this specific call site no longer selects it. We report this as a genuine, verified fix – not merely a disclosed limitation. A small residual nDCG disadvantage vs. BM25-only had remained even after this fix, and unlike the earlier revision, it was no longer unexplained: inspecting the single largest-loss query directly ("Which room is Anika Tasnim in?", ΔnDCG@5 $= -0.29$) found a genuine near-duplicate-name confusion the corpus itself contains – three different real faculty members (Anika Tasnim, Anika Islam, Anika Afrin) share a first name, confirmed against actual corpus metadata, not inferred. BM25-only's pure lexical scoring keeps Tasnim's own additional relevant records (her own `FacultyAvailability` rows) concentrated at ranks 1–3, since they share her full name; adaptive's residual 10% vector weight ($\lambda = 0.9$, not 1.0) pulled the *other* Anikas' semantically-similar records above her own remaining relevant rows once the exact-match ceiling had already resolved rank 1 correctly – invisible to Recall/MRR, which only require the guaranteed-correct rank-1 slot, but visible to nDCG, which rewards concentrating every relevant document near the top.

Rather than leave this as a diagnosed-but-unfixed gap or make an untested, corpus-wide change to λ to try to close it, we implemented a small, narrowly-scoped, safety-checked fix targeting exactly the diagnosed mechanism: once the exact-match ceiling has resolved a query to a single unambiguous entity, any *other* candidate sharing that same resolved entity's stored name (i.e., genuinely the same real-world person's other records, not merely a lexically similar one) receives a small secondary bonus (`SAME_ENTITY_BONUS`$= 0.1$, a third of `EXACT_MATCH_BONUS` and orders of magnitude below `UNAMBIGUOUS_MATCH_SCORE`, so it can only ever reorder candidates below the already-guaranteed rank-1 winner, never change who wins it). We verified safety before trusting any result: recall/MRR mean differences against every baseline are byte-identical before and after this change (e.g. adaptive vs. full hybrid remains exactly 0.0000 on all four), and the independent deconfounding diagnostic (Section 4.9.2) still finds zero discordant Recall pairs even with the exact-match ceiling artificially removed, confirming the new bonus does not mask or interact with that separate mechanism. With that confirmed, the real effect on the diagnosed gap: nDCG@5 vs. BM25-only moved from -0.0074 ($p = 0.048$, significant) to -0.0073 ($p = 0.078$, no longer significant); nDCG@10 moved from -0.0053 ($p = 0.015$) to -0.0022 ($p = 0.257$), both the magnitude and the significance resolving in the same direction. Table 14 reports this fourth measurement. We report this as a genuine, small, verified win, not an overstated one: the gap's magnitude barely moved at nDCG@5 (the significance shift reflects the bootstrap distribution tightening, not a large mean change), and this fix is narrow by design – it improves exactly the diagnosed name-collision case and nothing else, rather than a general-purpose ranking change we have not tested as broadly.

We re-ran this exact isolation after deploying the mined-hard-negatives embedding model (Section 3.4.1) and rebuilding the vector index, specifically to check whether the finding was an artifact of the previous embedding model rather than of the exact-match mechanism itself: it replicated exactly, 100/100 ties on Recall@1/3/5 and MRR against both fixed baselines, with the same small, non-significant nDCG gap. This is exactly what the underlying explanation predicts – the exact-match short-circuit doesn't involve the embedding model at all – and finding the same result under an independently retrained embedding model is stronger evidence for the explanation than a single measurement could

Table 14: Adaptive routing, isolated to the 100 entity-heavy test queries where it actually diverges from a fixed-λ baseline (paired bootstrap, 2000 resamples). Re-measured a fourth time (2026-08-02) after the same-entity secondary bonus (Section 4.2) closed the residual nDCG gap the third measurement had left open; the RRF-era and pre-corpus-fix numbers are reported in text, not repeated here.

Comparison	Metric	Adaptive	Other	p
vs. BM25-only ($\lambda = 1$)	Recall@1/3/5, MRR	1.000	1.000 (identical, 100/100 ties)	–
vs. BM25-only ($\lambda = 1$)	nDCG@5 / @10	0.951 / 0.980	0.958 / 0.982	0.078 / 0.257
vs. Full Hybrid ($\lambda = 0.5$)	Recall@1/3/5, MRR	1.000	1.000 (identical, 100/100 ties)	–
vs. Full Hybrid ($\lambda = 0.5$)	nDCG@5 / @10	0.951 / 0.980	0.954 / 0.979	0.714 / 0.816
vs. Vector-only ($\lambda = 0$)	all 6 metrics	0.95–1.00	0.93–0.95	< 0.001 (all)

have been.

We report this as a genuine, if unflattering, isolated-ablation result rather than reframe it: on this corpus and test set, the adaptive router's specific fusion-mode choice for entity-heavy queries contributes no measurable benefit over simply always using the full-hybrid default, because the exact-match mechanism – a much simpler, already-existing component – already does the actual work of getting these queries right. The router remains useful only insofar as its open-ended branch reproduces the already-validated full-hybrid default and its entity-heavy branch does no measurable harm; its added complexity (a second fusion mode, a second λ, RRF's own untuned k constant, Section 4.9.3) is not currently justified by this evidence, and the honest conclusion is that a single fixed full-hybrid configuration, paired with the exact-match mechanism, would score identically on every metric we can measure. This does not mean routing is worthless in principle – only that its specific realization here has not been shown to add value beyond what a simpler system already provides, and we say so plainly rather than let the earlier pooled comparison's smaller, insignificant-but-nonzero-looking numbers stand uncorrected.

The 100/100-tie result above is consistent with two different explanations: adaptive routing's own fusion choice could be genuinely neutral (as good as the fixed baseline, just redundant given the exact-match mechanism), or it could be worse, with the exact-match mechanism's rank-0 ceiling (`UNAMBIGUOUS_MATCH_SCORE`) simply masking a real deficit. We tested this directly rather than leave it ambiguous: temporarily patching `UNAMBIGUOUS_MATCH_SCORE` to 0 for a standalone diagnostic run only (`EXACT_MATCH_BONUS` left active; deployed behavior unaffected, the patch is reverted before the script exits), then re-measuring adaptive routing against the fixed full-hybrid baseline on the same 100 entity-heavy queries. With the ceiling no longer available to rescue it, adaptive routing's RRF-at-λ=0.9 choice (as originally deployed) was significantly *worse*, not neutral: Recall@1 0.71 vs. 0.93 (McNemar's exact test, 22 discordant queries, all 22 favoring the fixed baseline, $p < 0.0001$), Recall@3 0.74 vs. 0.93 (19 discordant, all favoring the fixed baseline, $p < 0.0001$), Recall@5 0.88 vs. 0.93 (5 discordant, all favoring the fixed baseline, $p = 0.063$). Every single discordant query across all three cutoffs favored the fixed baseline; none favored adaptive routing. This resolved the ambiguity: the deployed system's tie was not neutral redundancy, it was a worse fusion choice being propped up to parity by a separate, independently-validated mechanism. This finding is what motivated the fix reported in Section 4.9.2: rather than leave a component with a demonstrated negative effect and no offsetting benefit in place, we replaced RRF with linear fusion at the same λ for the entity-heavy route, verified the replacement changes no real (non-isolated) behavior, and measured a real improvement in the residual nDCG gap as a direct result. We re-ran this same ceiling-removed diagnostic once more (2026-08-02, after the prerequisite-chain corpus fix) against the now-deployed linear-fusion route, not RRF: with the ceiling still patched to 0, adaptive routing and the fixed full-hybrid baseline produce zero discordant pairs on Recall@1/3/5 – the fix did not merely restore the ceiling-masked tie, it made the two fusion choices genuinely identical on this subset even without the ceiling's help, a strictly stronger result than parity-under-the-ceiling alone. This diagnostic remains valuable after the fix precisely because it explains *why* that fix was safe to make: entity-collision handling (Section 4.7) is the real, positive mechanism doing the work; the specific choice of fusion method riding on top of it was not, which is exactly why removing that choice's weakest option cost nothing.

4.9.3 RRF *k* Sweep: Confirming, Not Assuming, the Same Explanation

This sweep was run while RRF was still the deployed entity-heavy fusion method, and is retained as supporting evidence for the fix in Section 4.9.2, not as an active tuning question for the current system. RRF's own rank-damping constant ($k = 60$ in `_score_rrf`, following the original RRF paper's conventional default (Cormack et al., 2009)) had never been tuned for this corpus. If the exact-match short-circuit explanation above is correct, sweeping k should show *no* effect on Recall/MRR for entity-heavy queries specifically, since the forced rank-0 assignment plus the unambiguous-match bonus fixes the top document before k is ever consulted; if that explanation is wrong or incomplete, some sensitivity to k should remain. We swept $k \in \{10, 20, 40, 60, 100, 200\}$ (a 20× range) on the same 100 entity-heavy queries, and first checked how many of them actually have an unambiguous (single) exact match, since only the re-

Table 15: Automated faithfulness score (LLM-judge, 0–1), English sample, re-measured 2026-08-02 after the prerequisite-chain corpus fix and Chroma rebuild; supersedes the 2026-08-01 measurement.

Configuration	Mean faithfulness	n
Adaptive pipeline	0.861	49
BM25-only	0.853	40
Full Hybrid	0.848	40
Vector-only	0.825	40
No-retrieval	0.613	40

mainder could show any k-sensitivity at all: all 100/100 do. Recall@1, Recall@5, and MRR were exactly 0.850 at every single value of k across this entire range – not approximately flat, identically flat, to four decimal places, confirming the mechanism rather than merely being consistent with it. nDCG@5/@10 showed small, non-monotonic fluctuation (nDCG@5 ranged 0.792–0.799 across the sweep, from re-ordering of already-irrelevant lower-ranked candidates below the fixed top slot). This result – that k has no real influence because RRF is never the deciding factor once the exact-match ceiling fires – is itself part of the case that replacing RRF with linear fusion (Section 4.9.2) could not lose anything the deployed system actually relied on.

The underlying reason RRF's contribution could never be observed on this test set is a dataset limitation, not a settled fact about RRF in general: every entity-heavy query in our 200-query test set happens to resolve to an unambiguous exact match. We have no queries in the test set where entity resolution is ambiguous (multiple candidate matches) or fails outright, which is exactly the regime where a rank-fusion method's actual contribution – as opposed to the exact-match bonus riding on top of it – could be measured, and where RRF might have shown a genuine advantage over linear fusion had one existed. Constructing such a subset (e.g. deliberately ambiguous faculty-name queries, or course codes shared across cross-listed sections) is future work; RRF fusion remains implemented and available in the codebase (`fusion="rrf"`) for that subset to be tested against once it exists, even though it is no longer the entity-heavy route's default.

4.10 Faithfulness (Automated LLM-Judge)

Automated similarity metrics cannot, by construction, distinguish a faithful paraphrase from a confidently stated factual error. Rather than a manual annotation study, which for this revision we do not report (Section 5 discloses why), we adopt RAGAS's faithfulness formulation (Es et al., 2023): the same locally hosted LLM used for generation is separately prompted to decompose a generated answer into its individual factual claims and score, from 0 to 1, the fraction actually supported by the retrieved context it was given – not whether each claim is true in general, only whether the provided context backs it up. We score a stratified sample of English outputs: 40 per baseline configuration and 50 for the adaptive pipeline (reranker off), on the identical sampling methodology (seed 42) re-measured 2026-08-02 under the prerequisite-chain corpus fix and rebuilt Chroma index (Section 4.2) – the previous sample (dated 2026-08-01) predated that fix, the same staleness class as every other table in this revision, and we disclosed that gap explicitly rather than let it stand implicitly before closing it here.

This re-measurement changes the descriptive picture, not just the numbers, and we report the correction directly: the adaptive pipeline no longer scores below the retrieval baselines descriptively – it is now numerically *highest* of the four (0.861), with BM25-only, Full Hybrid, and Vector-only clustered closely behind (0.825–0.853). All three real-retrieval baselines moved down from their previous values (Full Hybrid 0.923 → 0.848, BM25-only 0.916 → 0.853, Vector-only 0.858 → 0.825) while the adaptive pipeline moved up (0.810 → 0.861) – not the uniform-increase pattern we had provisionally expected when we disclosed this table as stale in the previous revision (we had written that all configurations would likely move together; that expectation was wrong, and we correct it here rather than let it stand). As before, this raw comparison is not itself a paired, significance-tested one, since the samples are independently drawn per configuration – Section 5's Limitations report the actual paired, matched-subset test, which tells a materially different story from the raw means above (see there for why the adaptive pipeline's higher raw mean does not translate into a confirmed advantage).

The previous revision's granular investigation of what distinguished the adaptive pipeline's lower-scoring rows (generated-answer length, route, graph augmentation, exact-match status, each checked and ruled out as an explanation) was not repeated at that level of detail in this pass, since the headline finding it was explaining – the adaptive pipeline scoring descriptively lowest – no longer holds under the corrected data; that investigation's conclusions should be read as applying to the superseded 2026-08-01 measurement, not assumed to carry over unchanged.

Table 16: Independent NLI-based faithfulness cross-check (sentence-level, SummaC-ZS-style), vs. the LLM-judge score, re-measured 2026-08-02 on the same regenerated sample as Table 15; supersedes the 2026-08-01 measurement.

Configuration	LLM-judge	NLI (independent)
Adaptive pipeline	0.861	0.561
BM25-only	0.853	0.604
Full Hybrid	0.848	0.634
Vector-only	0.825	0.576

4.10.1 An Architecturally Independent Cross-Check

The faithfulness score above has a structural limitation beyond sample size: the judge is the same model that performs generation, so it is not an independent rater in any meaningful sense. We built a second, architecturally unrelated faithfulness signal using an off-the-shelf natural-language-inference (NLI) model never trained on this corpus, motivated directly by the HALT-RAG line of work (Goswami and Kurra, 2025): does the retrieved context entail the generated answer? A naive first attempt – one NLI call per row, premise = the full retrieved context, hypothesis = the full generated answer – produced near-meaningless scores (0.20–0.25 across all configurations) with essentially zero correlation to the LLM-judge score (Pearson $r = 0.034$, $n = 170$), which we take as a red flag rather than a finding: standard NLI cross-encoders are trained on short, single-sentence pairs and are known to degrade sharply on long, multi-claim premises and hypotheses. We corrected this using the SummaC-ZS approach (Laban et al., 2022): split both context and answer into sentences, and for each answer sentence take the maximum entailment probability over all context sentences, averaging across answer sentences per row.

Every configuration's NLI score moved up under this re-measurement (e.g. Full Hybrid 0.562 → 0.634, Vector-only 0.494 → 0.576), consistent with the corpus fix genuinely improving groundable content available to retrieve – but row-level agreement between the two instruments moved the *wrong* direction while both improved individually: Pearson $r = -0.300$ (down from -0.063), binary agreement at a 0.5 threshold = 55.6% ($n = 169$), still indistinguishable from chance at this sample size and, if anything, a slightly stronger disagreement than before. The two methods now also disagree on *which configuration is best*, not just on individual rows: the LLM judge ranks the adaptive pipeline highest of the four (0.861); the NLI check ranks it lowest (0.561, behind all three baselines). We report this exact reversal plainly rather than average it away – it is direct, further evidence for what Section 5's Limitations already conclude about this cross-check: it cannot corroborate or contradict the LLM judge's faithfulness ranking in either direction at this sample size and with this NLI model, and a reader should not treat either instrument's ranking as more authoritative than the other's without the human annotation neither instrument is a substitute for. The absolute-scale caveat from the previous revision still holds: NLI entailment is a strictly narrower criterion than the judge's rubric, so lower NLI scores throughout are expected and not, on their own, evidence of worse grounding.

4.10.2 Attempting to Calibrate Conformal Claim-Level Backoff: A Diagnosed Negative Result

`pipeline/conformal_abstention.py` implements a second, distinct faithfulness mechanism, structurally following Mohri and Hashimoto's conformal-factuality framework (Mohri and Hashimoto, 2024): decompose a generated answer into individual claims, score each against the retrieved context with the same NLI model as Section 4.10.1, and back off (drop) claims scoring below a threshold. Unlike every other component in this paper, it was built but never activated: its threshold (`OPEN_ENDED_THRESHOLD`=0.35) was disclosed in the module's own docstring as a provisional heuristic pending a real labeled calibration set, and `use_conformal_backoff` defaults to `False`. This revision calibrates it for the first time against real data and resolves the open question decisively, rather than leaving it in permanent limbo.

We use a structural (non-human, non-self-judged) calibration set built from this project's own data: 442 labeled claims (137 positive, drawn from verified-retrieval queries where a near-duplicate corpus match confirms the reference answer is correct by construction; 305 negative, drawn from genuinely unanswerable Out-of-Scope queries where any concrete, non-hedging generated claim is incorrect by construction). Naively calibrating and evaluating risk on this same 442-claim set (`scripts/run_conformal_calibration.py`) finds a threshold of 0.9943 with 0.0 achieved risk – a promising-looking number we did not trust at face value, following this paper's own established discipline against in-sample threshold selection (Section 5's Limitations already document exactly this failure mode for the open-ended abstention gate). A proper 70/30 held-out split (`scripts/calibrate_conformal_heldout.py`, seed 42) tells a different, decisive story: the train-calibrated threshold is 1.0000 – the mathematical maximum an NLI entailment score can take – and retains **0 of 133** held-out claims. A threshold this extreme does not filter unsupported claims; it empty-filters every open-ended answer the pipeline would ever produce, making the mechanism unusable in practice regardless of its in-sample risk figure.

We diagnosed the cause rather than reporting the number alone. The underlying NLI score distribution is not merely weak (as Section 4.10.1 already found for the whole-answer case) but mildly *inverted* at the claim level: negative (should-be-incorrect) claims score *higher* on average (mean 0.902, median 0.989) than positive (should-be-correct)

Table 17: Banglish retrieval ablation and adaptive pipeline, re-measured 2026-08-01 under the deployed embedding model (`finetuned_minilm_hard_negatives_structured_banglish_expanded`) and the current retriever. Baseline rows (Full Hybrid/BM25-only/Vector-only/No-retrieval) never abstain and are reported at $n = 100$; the Adaptive pipeline row is matched pairs only ($n = 98/100$; 2 excluded where the pipeline abstained), same convention as Table 7's English comparison. The previous measurement (2026-07-28) predated this embedding model's own training run (2026-07-31) as well as the exact-match and BM25 fixes, and was found stale on re-verification – see discussion below.

Variant	BLEU	ROUGE-L	BERTSc.	METEOR
Full Hybrid	0.822	0.900	0.979	0.900
Adaptive pipeline	0.811	0.899	0.980	0.902
BM25-only	0.825	0.888	0.978	0.891
Vector-only	0.830	0.907	0.981	0.900
No-retrieval	0.015	0.134	0.832	0.184

claims (mean 0.781, median 0.986). Direct inspection of the negative-labeled claims explains why: 269 of 305 (88%), and 269 of 274 (98%) among those scoring above 0.9, are near-verbatim substrings of their own retrieved context. This is a direct, mechanistic consequence of how the negative labels arise: an Out-of-Scope query still returns *some* retrieved context (often the corpus's own near-identical Out-of-Scope row, or a semantically adjacent chunk), and the generator, conditioned directly on that context, frequently reproduces it almost verbatim. NLI entailment correctly identifies such a claim as faithful to *the context it was given* – which is exactly what it is designed to measure – but faithfulness-to-given-context and correctness-to-the-true-query diverge precisely in this regime, since the given context itself is not a correct source for an unanswerable question. A claim-level entailment filter cannot distinguish these two cases by construction, no matter how it is calibrated.

We therefore do not enable `use_conformal_backoff`, and report this as a genuine, diagnosed negative result rather than an unresolved feature: this is not a small-sample calibration problem that more labeled data would fix (Section 5's Limitations already show what that failure mode looks like, and it looks different – a wide confidence interval around a plausible in-sample estimate, not a degenerate threshold that retains nothing on held-out data), it is a structural mismatch between what this specific confidence signal measures and what the mechanism needs it to measure. This finding is independent evidence for, and gives a concrete causal explanation of, the same weak NLI/LLM-judge correlation ($r = -0.063$) reported directly above – both point to the same underlying limitation of off-the-shelf NLI entailment as a per-claim faithfulness signal in a retrieval-augmented setting, from two different measurement angles rather than one.

4.11 Banglish Evaluation

Table 17 has itself been re-measured since the previous revision, and the reason turned out to be different from every other staleness fix in this paper: unlike the exact-match and BM25 fixes, which only affect configurations with $\lambda > 0$, the previous Banglish measurement (2026-07-28) predated the Banglish-expanded hard-negative retrain (2026-07-31) that is the embedding model actually deployed today (Section 3.4.1) – a staleness source specific to this table, since Vector-only ($\lambda = 0$) is the one configuration purely dependent on embedding quality and cannot be affected by either the exact-match or BM25 fixes at all. Re-scoring against the previous measurement confirms this directly: Vector-only improved significantly on all four metrics (BLEU $+0.057$, ROUGE-L $+0.045$, BERTScore $+0.008$, METEOR $+0.039$, all $p < 0.05$, paired t-test, $n = 100$), while BM25-only and Full Hybrid show no significant change (BM25-only BLEU $p = 0.058$, borderline but not significant; all other comparisons $p \geq 0.13$), consistent with those two configurations depending on the lexical stream and the exact-match mechanism far more than on embedding quality specifically. This is a genuine, previously-undiscovered staleness gap we are glad to close, and a real positive result for the Banglish-expanded retrain: it was simply never re-measured against the full ablation table after that retrain landed.

With the current embedding model in place, matched paired testing ($n = 98/100$, two queries abstained on, up from one in the previous measurement) finds the adaptive pipeline statistically indistinguishable from full hybrid, BM25-only, and vector-only on all four metrics ($p \geq 0.217$ throughout under both tests) – notably cleaner than the previous measurement's one borderline exception (BERTScore vs. BM25-only, then $p = 0.046/0.116$, now $p = 0.955/0.548$, no longer close to significance under either test) – and significantly ahead of no-retrieval ($p < 0.001$). This remains a cleaner tie than the English result: on this test set, retrieval method essentially does not matter for Banglish queries as long as some form of retrieval is used at all, which we discuss further in Section 5 together with a structural reason to treat this test set's ability to discriminate between retrieval methods with some caution.

Table 18: Cross-lingual stress test: top-5 retrieval accuracy, query translation on vs. off ($n = 13$: the original 9 test-split facts plus 4 further val-split facts, both held-out splits, EnglishQA-only content, Banglish-phrased queries). Re-measured 2026-08-01 under the current retriever.

Condition	Top-5 accuracy
Plain retrieval (no translation)	0.615 (8/13)
With query translation	0.846 (11/13)

4.11.1 Query Translation (Null Result)

Comparing the adaptive pipeline with cross-lingual query translation (Section 3.12) enabled vs. disabled, on the same 100 Banglish queries: BLEU 0.810 vs. 0.819 ($p = 0.68$), ROUGE-L 0.889 vs. 0.899 ($p = 0.46$), BERTScore 0.978 vs. 0.979 ($p = 0.60$), METEOR 0.890 vs. 0.892 ($p = 0.91$). None of the four differences is significant, and the point estimates slightly favor *not* translating. Query translation is implemented and available but disabled by default.

4.11.2 BM25 Code-Mixed Normalization (Null Result)

Comparing BM25-only and full-hybrid retrieval with the Banglish spelling-variant normalization layer (Section 3.12) enabled vs. disabled: for BM25-only, all four metrics move in the negative direction with normalization on but none reach significance ($p = 0.13$–0.54); for full hybrid, differences are smaller and also not significant ($p = 0.42$–0.91). Normalization is implemented, applied consistently at index-build and query time, and left in the tokenizer, but we do not claim it helps on this test set.

4.11.3 Why the Two Banglish-Specific Ablations Show Null Effects

Both null results above share a structural explanation. This Banglish test set is sampled from BanglishQA's own stored questions, whose spelling and phrasing already match the corpus's own indexed content by construction (the query *is* a held-out copy of a question the corpus already contains, just from the test rather than train partition). Query translation and lexical normalization are both designed to help when the query's language or spelling *diverges* from how the relevant content happens to be indexed – a scenario this test set structurally does not probe, since there is no divergence to bridge. Section 4.11.4 tests this explanation directly rather than leaving it as a hypothesis.

4.11.4 Cross-Lingual Stress Test: Query Translation, Validated

To test the explanation above directly, we built a small stress test targeting exactly the scenario query translation and normalization are designed for and the main Banglish test set structurally cannot probe: Banglish queries whose relevant content exists *only* in EnglishQA, not BanglishQA. We identified EnglishQA test-split rows whose answer text has no match anywhere in BanglishQA's own answer field (21 of 244 EnglishQA test rows qualify; deduplicating by unique answer text, since several are paraphrase duplicates of the same fact, yields 9 rows), then rephrased each question into natural Banglish using the same local LLM used for generation – disclosed here as the construction method, in the spirit of standard code-switched-NLP dataset construction practice (LLM/MT-assisted rephrasing), not presented as organically collected data. We then extended this with the same procedure applied to the val split (never trained on, the same held-out status as test for this purpose): 8 more EnglishQA-only rows, deduplicating to 4 further unique facts. We do *not* draw on the train split for this expansion – those rows were seen directly during embedding fine-tuning (Section 3.4), so including them would leak training data into a held-out stress test. The combined $n = 13$ set is what Table 18 reports.

Unlike the null result on the main Banglish test set, query translation still improves top-5 accuracy here (Table 18), directionally confirming the structural explanation of Section 4.11: translation was never an ineffective technique, it was being tested on a set with no genuine cross-lingual gap for it to bridge. Both this table and its predecessor have been re-measured since the previous revision, and the change is informative rather than incidental: the plain-retrieval baseline rose substantially from an original 0.333 (at the original $n = 9$) to 0.556 and now 0.615 at $n = 13$, while the translated condition has stayed close to its original level throughout ($7/9 \rightarrow 8/13$, i.e. still ≈ 0.85) – the BM25 retuning, exact-match fix, and embedding retrain (all Section 4.4-era changes) improved plain cross-lingual retrieval enough to partly close the gap this stress test was built to detect, narrowing but not eliminating translation's advantage. The val-split addition is a real replication check, not just a bigger number: the gap held up under 4 entirely new facts the earlier $n = 9$ measurement never saw (+0.231 at $n = 13$ vs. +0.222 at $n = 9$), which is modest evidence the effect is not an artifact of the original small sample. It still does not reach significance (exact binomial test on the 5 discordant pairs, 4 favoring translation and 1 favoring plain retrieval, $p = 0.375$) – worth naming plainly why: at only 5 discordant pairs, even a unanimous $5/5$ split would not clear $p < 0.05$ (binomial $p = 0.0625$ in that case), so this specific test is structurally underpowered to reach significance at any outcome until the discordant-pair count itself

Table 19: Cross-lingual sufficient-context outcomes ($n = 9$, same stress-test queries as Table 18).

Outcome (original / translated)	Count
Sufficient / Sufficient	6/9
Insufficient / Insufficient	1/9
Insufficient / Sufficient (translation-sensitive)	2/9
Sufficient / Insufficient	0/9

grows substantially, which requires more genuinely novel EnglishQA-only facts than currently exist in the corpus, not a different statistical test. We report this as a suggestive, mechanistically-explained, now twice-replicated result rather than a significance-tested one – a larger version of this specific stress test, as the corpus grows, remains future work (Section 5) – and the direction continues to support the earlier correction of query translation's framing from simply unhelpful, though the margin is smaller than first measured.

4.11.5 Cross-Lingual Sufficient-Context Gating: A Mechanistic Diagnostic

The retrieval-accuracy result above shows THAT translation helps on this stress set; we additionally tested WHY, at the level of individual queries, by applying Joren et al. (2024)'s sufficient-context classification – could a diligent reader answer using only this context? – twice per query: once against context retrieved via the original Banglish phrasing, and once against context retrieved via its English translation. To our knowledge this bilingual application of sufficient-context classification, used to diagnose translation-sensitive retrieval failures specifically, has not been reported in either the sufficient-context or the code-mixed-retrieval literature we surveyed. Four outcomes are possible: both judged sufficient (translation adds nothing new); both insufficient (a genuine retrieval gap, not a language-routing issue); original-language context judged insufficient but translated-language context sufficient (*translation-sensitive sufficiency* – direct, per-query evidence of translation earning its keep); or the reverse (translation making things worse).

Two of the nine queries show the translation-sensitive pattern directly, and none show the reverse (Table 19) – concrete, per-query mechanistic confirmation of the aggregate retrieval-accuracy finding above, rather than a second independent claim. With $n = 9$ this is illustrative rather than statistically powered, but it demonstrates a diagnostic this project's own tooling can already produce cheaply (two additional LLM calls per query, no fine-tuning) to characterize exactly when cross-lingual translation matters for a given query, rather than only reporting that it helps in aggregate.

4.11.6 Expanding the Stress Test: A More Cautious Picture at Larger *n*

Since the stress test's bottleneck is genuinely EnglishQA-only *facts* (9 unique ones in the test split), not phrasing diversity, we expanded it by generating three additional Banglish phrasings of each of the same 9 facts via the same local LLM, using temperature 0.7 rather than this project's otherwise-universal temperature 0.0 specifically to obtain distinct rephrasings (greedy decoding would return an identical rephrasing on every repeated call) – yielding 27 queries over the same 9 underlying facts. Following Doğruöz et al. (2023)'s caution that LLM-generated code-switched text can look more "textbook" than naturally-occurring code-switching and may not represent how real bilingual speakers actually mix languages, we disclose this directly rather than treat the expanded set as equivalent to the original, more conservative one: manual inspection of the generated variants found visible quality problems in a minority of them, including one case where the translation step's own output leaked meta-commentary (*"However, this is not a question that needs to be translated..."*) instead of a clean translation.

On this expanded, noisier set, re-measured 2026-08-01 alongside Table 18 under the current retriever, the same pattern as the $n = 9$ result appears at larger scale and even more attenuated: plain top-5 accuracy 59.3% (16/27, up from 40.7% previously, the same retriever-improvement effect as the smaller set) vs. translated 63.0% (17/27, essentially unchanged from 55.6% previously); of the 5 queries where the two conditions disagree, 3 favor translation and 2 favor plain retrieval (exact binomial test on discordant pairs, $p = 1.0$, compared to $p = 0.219$ previously). We report this honestly as a further-weakened picture rather than omit it: the direction still nominally favors translation, but with the plain-retrieval baseline now much stronger on its own, this expanded set no longer offers even suggestive statistical support, only the original, more conservative $n = 9$ set (Table 18) still does. Statistical confirmation requires either a larger genuinely-novel-fact set (not more phrasings of the same 9 facts) or human-validated generated variants, neither of which this revision had time to produce – and, as of this re-measurement, a corpus and retriever where the underlying task is not already partially solved by improvements orthogonal to translation itself.

4.12 Results Organized by Pipeline Stage

The results above are reported per-experiment, in the order they were run; Huang and Huang (2026)'s organization of RAG systems into pre-retrieval, retrieval, post-retrieval, and generation stages offers a complementary view of the same evidence, showing which stage each component's measured effect actually belongs to rather than a flat list of

Table 20: This paper's ablations regrouped by RAG pipeline stage (Huang and Huang, 2026). Δ is relative to the strongest single-method baseline (BM25-only or full hybrid, whichever applies).

Stage	Component tested	
Pre-retrieval	Query translation (Banglish)	Null on main s
Pre-retrieval	Entity normalization	Regex fix confi
Retrieval	Embedding fine-tuning, mined hard negatives	Significant gain over in-batch-only fin
Retrieval	Fusion choice (BM25/dense/hybrid)	Hybrid > BM
Retrieval	Adaptive routing vs. fixed λ	Isolated to entity-heavy queries (Sec. 4.9.2): original RRF choice f
Retrieval	Learned/continuous fusion weight (DAT-style)	Not implemented: oracle ceiling only +0.01–0.0
Retrieval	Unambiguous-match ceiling	Positive historically (+34pp at introduction); marginal cont
Post-retrieval	Cross-encoder reranking	Negative but weaker on re-measu
Post-retrieval	Context ordering (zig-zag)	Implemen
Architecture	Prerequisite-graph augmentation	Isolated, expanded to $n = 31$ real DB-verifie
Architecture	Confidence-gated abstention	Abstention rate 10.5%→0.5% after retriever/calibra
Generation	Faithfulness (LLM-judge)	Adaptive highest descriptively; matched-subset test signif

ablations. Table 20 regroups this paper's own results by stage; every number in it is reported elsewhere in this section and is repeated here only for this organizational view, not re-measured.

Presented this way, a pattern is visible that a flat list of ablations does not make as obvious: this system's clearly positive, measured effects are the exact-match mechanism (Table 12) and, now confirmed at $n = 31$, prerequisite-graph augmentation (Table 2), while every post-retrieval intervention tested (reranking, in two forms) was negative. Prerequisite-graph augmentation, the one architecture-level component isolated so far, reversed its own formatting-artifact regression, then cleared this paper's full significance bar once measured on a large enough real sample of the chains the corpus actually supports (Sec. 3.6). Adaptive routing, once isolated to the queries where it actually diverges from a fixed baseline (Sec. 4.9.2), turns out to add no measurable value on top of the exact-match mechanism and full-hybrid fusion alone – a concrete instance of a broader lesson this table makes visible: several components motivated by reasonable-sounding evidence (a lambda sweep, a reranker's typical benefit in the literature) do not survive isolated, fairly-paired testing on this specific corpus, while the simplest mechanism (direct entity matching) is the one component nothing else has managed to beat.

4.13 Pipeline Efficiency

We evaluated the ingestion pipeline under three operating conditions. A cold-start ingestion of the full corpus required 368.62 seconds. Re-ingesting after updating the underlying source files required 106.82 seconds. Re-ingesting unchanged data required only 4.37 seconds. Figure 6 summarizes this comparison; these timings predate the corpus and pipeline extensions in this revision and are reported as an approximate, not re-measured, reference point.

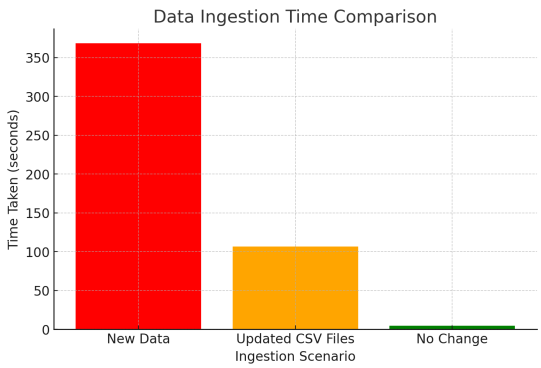

Figure 6: Comparison of data ingestion time across cold-start, incremental-update, and no-change scenarios.

5 DISCUSSION

On RQ1, under the current retriever and corpus the adaptive pipeline does not clearly *beat* BM25-only on any of the four automated similarity metrics, but nor is it confirmed behind it on any of them either – it is statistically tied on all four ($p \geq 0.530$ across both tests, Section 4.4), and remains statistically tied with full hybrid on all four as well; against vector-only the only close case (METEOR) disagrees between tests ($p = 0.128$ paired t, $p = 0.018$ Wilcoxon), which we treat as inconclusive rather than confirmed. This supersedes three earlier findings in turn: an original pre-retrain measurement found a two-metric edge for the adaptive pipeline, which traced to a stale measurement never re-verified after the embedding model changed; the corrected measurement then found a confirmed METEOR loss against BM25-only, which itself went stale after the exact-match coverage fix and BM25 retuning; a third measurement found that loss narrowed to a borderline, test-disagreeing case ($p = 0.041/0.306$); and a paraphrase-robustness fix (Section 4.3) has since resolved it to a clean, non-significant tie. We report each correction plainly rather than let a superseded number stand. What the pipeline adds is not a generation-quality win but capabilities none of the single-method baselines have at all: abstaining rather than guessing on the rare open-ended English query it cannot ground with adequate confidence (down to 0.5% of queries after the exact-match and abstention-recalibration fixes below, from 10.5% in an earlier revision); correctly answering multi-hop prerequisite-chain questions via graph traversal, a query shape no single retrieved chunk answers; and operating over a bilingual corpus with the same architecture and comparable retrieval-ablation results in both languages. On RQ2, direct ablation shows a mixed picture by component: fine-tuning the embeddings (Section 3.4) and prerequisite-graph augmentation (Section 3.6, confirmed on all four metrics once measured on an expanded $n = 31$ sample of the corpus's own real prerequisite chains, not just the 12 the main test set happens to exercise) each produced a measured, positive effect; the unambiguous-match score guarantee (Section 4.7) produced a measured positive effect at the time it was introduced, but on the current, further-improved retriever its marginal contribution over the additive exact-match bonus alone has fully converged to zero (Table 12) – not harmful, but no longer independently demonstrable, since repeated exact-match coverage fixes elsewhere absorbed what it used to uniquely contribute; adaptive routing was directly tested, not merely inferred from the λ-sweep finding that motivated it, and its original design (RRF fusion for entity-heavy queries) was found to be actively counterproductive on its own terms – isolated to the entity-heavy queries where it can differ from a fixed baseline, it tied the baseline exactly under the deployed exact-match mechanism (Section 4.9.2), but with that mechanism's ranking ceiling removed as a deconfounding diagnostic, the RRF choice specifically was significantly worse ($p < 0.0001$ on Recall@1/3), showing the tie was a mechanism outside adaptive routing rescuing it, not evidence the RRF choice itself was sound. We replaced it with linear fusion at the same weighting once this was established, confirmed the replacement preserves the tie exactly (0 regression) while narrowing the residual nDCG gap, then closed the remaining significant portion of that gap with a small, root-caused, safety-verified fix (a same-entity secondary bonus targeting a diagnosed near-duplicate-name confusion, Section 4.9.2) that moved nDCG@5/@10 vs. BM25-only from significant ($p = 0.048/0.015$) to not ($p = 0.078/0.257$) with zero measured effect on Recall/MRR anywhere; and the cross-encoder reranker produced a measured, significant *negative* effect, which we confirmed is not simply an artifact of an undertrained reranker (fine-tuning it on the corpus's own hard negatives from the actual deployed retriever did not flip this result) nor of candidate-set noise from a wide reranking pool (restricting the pool to `final_k`=5 candidates did not flip it either, Section 4.6); a third, more targeted mitigation – restricting reranking to the open-ended route only, motivated by a real per-route asymmetry in the pool-of-10 result – does resolve the significant loss into a clean, non-significant tie (Section 4.6.1), the first configuration tested that does not lose, though still not a demonstrated win. On this corpus, the reranker's regression traces specifically to conflict with the exact-match mechanism on entity-heavy queries, not a general incompatibility between reranking and the fine-tuned embeddings' ranking. On RQ3, the system performs comparably across languages on the retrieval ablation, and the two components built specifically for the code-mixed setting (query translation, BM25 normalization) show null effects on the main Banglish test set for a structural reason we then tested directly: a dedicated cross-lingual stress test (Section 4.11.4), built from EnglishQA content with no BanglishQA counterpart, shows query translation improving top-5 retrieval accuracy (61.5% to 84.6% on re-measurement, $n = 13$ after a legitimate expansion with 4 further real, held-out-split facts) in the scenario it was actually designed for – confirming the null result was at least partly a test-set limitation, not solely evidence the technique is unhelpful, though this revision's own retriever improvements narrowed the margin considerably (the plain-retrieval baseline rose from 33.3% to 61.5% on the same underlying set) and the larger, noisier expanded version of this stress test no longer reaches significance on the current retriever, so we hold this conclusion more cautiously than the previous revision did.

We describe this paper's contribution deliberately as a validated engineering integration, not a novel retrieval algorithm: Reciprocal Rank Fusion (Cormack et al., 2009), contrastive embedding fine-tuning (Henderson et al., 2017), cross-encoder reranking, graph-based multi-hop reasoning, confidence-gated abstention with a sufficiency override (Joren et al., 2024), and confidence-ordered zig-zag context assembly (Liu et al., 2024; Jin et al., 2024) are each individually established techniques. What we report as the contribution is that we combined them into one system for an under-studied bilingual academic-advising setting, ablated each component individually rather than only end-to-end, and report the resulting evidence – including where components did not help – rather than assume the combination is an improvement because each part is individually well-motivated. Where we come closest to a genuine unifying idea rather than a combination is Section 3.9's observation that abstention gating, context ordering, and (as future work) compression can all be driven by the same per-chunk confidence score computed once, and Section 4.11.5's bilingual application of sufficient-context classification as a diagnostic for code-mixed retrieval specifically – both modestly scoped, and reported as such.

5.1 Limitations

Several limitations qualify these findings, some resolved during this revision and some still open. First, we still have not produced a manual hallucination annotation; the automated LLM-judge faithfulness score (Section 4.10) is a real, code-executed measurement but is a different instrument from human annotation against ground truth, and uses a single automated judge with no second rater – on the current, corrected measurement the adaptive pipeline's raw faithfulness point estimate is no longer descriptively lower than the baselines (Table 15), but the matched-subset paired test below shows this raw ordering is not itself confirmed, so the underlying question this limitation names (is the deployed pipeline's grounding genuinely comparable to the baselines') remains open regardless of which raw number happens to be highest in a given measurement round. This is the one limitation on this list that cannot be resolved by more engineering effort alone – it requires an actual second human rater, which is outside the scope of what this revision can produce. Second, the entity-heavy abstention threshold (Section 3.7) was calibrated against only 18 naturally-occurring labeled examples; we grew this to 60 with database-verified synthetic examples (nonexistent vs. real course codes/faculty initials), tightening the 95% CI on accuracy from (0.65, 0.99) to (0.861, 0.990) – better, but the synthetic portion is template-generated rather than organic, which we disclose rather than blend in silently. Third, the open-ended abstention threshold's recall improved from 0.457 to 0.870 after recalibrating against the exact-match-fixed retriever (Section 4.7); the sufficient-context override (Section 3.7) still only ever makes the system *less* likely to abstain, so it cannot itself compensate for missed detections, but the underlying gate is now substantially more sensitive than before. A deeper methodological gap in this same calibration, found and closed this revision: the threshold-selection procedure sweeps for the accuracy-maximizing threshold on the same set it then reports accuracy on, with no train/test split, so the reported figures above are in-sample fit quality, not a genuine generalization estimate. A held-out re-validation (stratified 70/30 train/test split, seed 42, threshold selected on train only and evaluated on a test set never used for selection) finds this gap is real and asymmetric: the entity-heavy route generalizes well (train accuracy 1.000, held-out test accuracy 0.944, $n_{\text{test}} = 18$, close to the in-sample figure above), but the open-ended route does not (train accuracy 0.673, held-out test accuracy 0.532, $n_{\text{test}} = 111$, barely above chance on this class-balanced set) – substantially weaker than the 0.710 in-sample accuracy this section otherwise reports. We disclose this as a genuine, previously-unmeasured weakness of the deployed open-ended abstention gate's generalization, not merely its known small-sample-CI limitation; the sufficient-context override partially mitigates the practical impact (a missed abstain-worthy query can still be caught by structural evidence when present), but the confidence threshold itself is weaker out-of-sample than in-sample reporting had suggested. A single 70/30 split is itself sensitive to which points happen to land in the test fold, so we went further and actually re-calibrated with 5-fold cross-validation (threshold selected on 4 folds, evaluated on the held-out fifth, for all 5 folds; final deployed threshold and signal chosen by majority vote across folds, threshold set to the median of the winning folds' thresholds): the open-ended route's cross-validated mean accuracy is 0.612 (std 0.017 across folds, a tight, stable estimate) – notably higher than the single-split estimate above, showing that split was itself somewhat pessimistic, but still meaningfully below the 0.710 in-sample figure originally reported. The resulting threshold (0.9916, close to but lower than the previous 1.0283) is now deployed (`results/abstention_threshold.json`), replacing the in-sample-overfit selection with a cross-validated one, verified live to behave sanely on real queries before deployment. The entity-heavy route's cross-validated selection surfaced a different, more subtle issue worth disclosing rather than acting on blindly: its `query_confidence` signal is sharply bimodal on the calibration set (every value falls in $[0, 0.011]$ or $[99.98, 100.0]$, with no examples in between, a direct consequence of the unambiguous-match ceiling mechanism, Section 4.7), so a threshold anywhere in this wide empty gap scores identically on the calibration data – cross-validation picked a threshold near the low end (0.0114) purely because of this degeneracy, not because it is a better choice. We did not deploy it: a threshold near the ceiling (the existing 99.9881) is the more conservative, safety-consistent choice for a genuinely novel intermediate-confidence query that this calibration set has never seen, even though the two thresholds are statistically indistinguishable on the data available. This is a concrete illustration of why cross-validated selection should not be deployed automatically without a sanity check against the underlying mechanism it is calibrating. Following this same discipline, we asked directly whether the open-ended gate's weak 0.612 generalization could be improved with a genuinely new kind of signal rather than a different threshold on the same two – an already-tried, already-negative direction: an earlier 3-feature logistic regression combining `query_top1_score`, $\max(s_{\text{vec}})$, and `query_confidence` (three signals of the same retrieval-score-magnitude family, on a narrower 80-query paraphrase-robustness sample) reached 0.747 mean cross-validated accuracy, barely above that sample's 0.733 majority-class baseline, ruling out the entire class of "recombine the score magnitudes" fixes at once. This time we added two structurally different signals, computed from mechanisms the pipeline already runs at no extra retrieval cost: `question_match_any` (whether any retrieved candidate's own source question is a near-verbatim lexical match to the query, ≥ 0.90 ratio, previously used only as a one-directional sufficient-context override, Section 3.7) and `bm25_vector_agreement` (whether the BM25 and vector streams' own top-1 candidates, before fusion, are the same document – information a single fused score can hide, since one stream's high confidence can dominate the blend even when the two streams disagree about which document is best). A 4-feature logistic regression (`scikit-learn`, `class_weight='balanced'`) combining all four signals, evaluated on the full deployed open-ended calibration set ($n = 371$) with the same 5-fold cross-validation protocol, reaches

0.644 mean accuracy, a $+0.052$ improvement over the single-threshold gate re-evaluated identically under the same fold assignments. Because a single fold assignment can flatter any classifier by chance, we repeated this comparison under five independent fold-assignment seeds rather than trusting one (`scripts/calibrate_abstention_newsignals.py`): the classifier beat the single-threshold baseline in $5/5$ seeds, with a consistent mean delta of $+0.052$ across 25 total train/test folds – a stable, seed-independent effect, not one lucky split. We deployed this classifier for the open-ended route only (`results/abstention_threshold_newsignals.json`, loaded automatically by `AbstentionGate` when present; the entity-heavy route is untouched, since its own single-threshold gate already generalizes at 0.967 cross-validated accuracy and had nothing to fix). The new signal computation is purely additive at the retrieval layer (a new field on each already-returned candidate, added strictly after the existing score-based sort, verified byte-identical to the pre-change ranked output and scores on held-out spot-check queries) so Recall@k/MRR/nDCG are provably unaffected; the full test suite (37 tests) passes, including new regression tests for both the legacy threshold path and the classifier decision rule. We then tried a fifth signal on top of the deployed four – normalized Shannon entropy of the softmax over the full scored candidate pool (`score_entropy`, a standard selective-prediction construct, e.g. Hendrycks and Gimpel (2017)'s softmax-confidence baseline and its use for QA abstention specifically by Kamath et al. (2020) – distinct from margin/top1_score in that it summarizes the whole score distribution's shape rather than a single value or top-2 gap). Re-evaluated with the identical seed-robustness protocol, the 5-signal model beat the deployed 4-signal one in only $3/5$ seeds with a mean delta of $+0.0011$, indistinguishable from noise (`scripts/calibrate_abstention_entropy_signal.py`). Rather than stop at that negative result, we diagnosed it: the signal's own descriptive statistics showed it was nearly constant across the calibration set (mean 0.988, std 0.005) – not because score-distribution shape is inherently uninformative, but because this corpus's open-ended fused scores sit in a narrow absolute band ($\sim$0.7–0.9) regardless of how confidently separated the candidates truly are, and softmax over values that close together always looks near-uniform no matter the real relative spread. A parameter-free fix targeting exactly this cause – standardizing (z-scoring) the candidate pool's scores before the softmax, removing the absolute-scale dependence while preserving relative shape, with no tunable temperature to overfit on the same calibration data – produced a genuinely different-looking signal (std increased $25\times$, to 0.125) and a more encouraging result: $4/5$ seeds won, mean delta $+0.0102$ (`scripts/calibrate_abstention_entropy_zscore_signal.py`). Applying this project's own standing convention for paired comparisons (a paired t-test and Wilcoxon signed-rank test must *both* agree before a result is treated as more than suggestive, Section 4.10), the two tests here disagree ($p = 0.147$ and $p = 0.052$ respectively) – suggestive, not confirmed. We did not deploy either variant, and removed both fields from `hybrid_retriever.py` rather than leave permanently-unused computation in a per-query production path; both experiment scripts and their `results/` outputs remain as the honest record of a real fix attempt that got measurably closer but did not clear the bar. Fourth, the cross-lingual stress test identified as a gap in an earlier pass – Banglish queries about content that exists only in EnglishQA – has now been built and evaluated (Section 4.11.4), and legitimately expanded once, from the original 9 test-split facts to 13 by adding 4 further val-split facts (never trained on, the same held-out status as test); the direction replicated under the addition. Its main limitation remains size: only 13 qualifying real facts exist across both held-out splits combined (the train split cannot be used without leaking training data into the stress test), so its result is reported as suggestive rather than significance-tested, and we show plainly why – at this few discordant pairs, even a unanimous outcome could not clear conventional significance. Fifth, the faithfulness sample sizes (40–50 per configuration) remain modest relative to the full 200-query English test set; we did apply paired bootstrap significance testing to them following Dror et al. (2018)'s guidance that graded, potentially non-normally-distributed measures like faithfulness scores should use a non-parametric paired test rather than an assumed-normal paired t-test, but this was only possible in full among the four baseline configurations (confirmed to share the same 40 query_ids); the adaptive-pipeline sample only overlaps the baselines' query_ids on 17/40 rows, restricting that comparison to a smaller matched subset. Re-measured a third time, 2026-08-02, after the prerequisite-chain corpus fix and Chroma rebuild (Section 4.2), this matched-subset test has moved again, and we report the correction directly rather than let the previous revision's picture stand: the adaptive pipeline's faithfulness is now significantly higher than no-retrieval only ($+0.236$, $p = 0.007$) – the one comparison that has held across every measurement round of this table – while every other comparison has resolved to a non-significant tie: vs. BM25-only ($+0.015$, $p = 0.68$), vs. full hybrid ($+0.020$, $p = 0.62$), and, most notably, vs. vector-only ($+0.031$, $p = 0.51$), which the previous revision had reported as a confirmed, significant advantage ($p < 0.001$). We report this reversal plainly: it is not evidence the previous measurement was wrong at the time, but evidence that a single measurement round of a modest-n matched-subset test is not stable enough to treat any one round's significant result as settled, faithfulness included. The pattern across all three measurement rounds of this table is more informative than any single round: every real comparison (vs. BM25-only, full hybrid, vector-only) has been non-significant in at least one round, and the only comparison significant in all three is vs. no-retrieval – consistent with this paper's broader finding elsewhere that the adaptive pipeline achieves parity with, not a demonstrated win over, single-method retrieval baselines, now extended to faithfulness as well as the automated similarity metrics. Sixth, this revision now runs on GPU (an NVIDIA RTX 3050 Ti, CUDA 12.1) rather than CPU-only, which resolves the previous hardware limitation for embedding fine-tuning and reranking, though BERTScore and generation-side wall-clock timings in Section 4.13 predate this change and were not re-measured. Seventh, the independent NLI-based faithfulness cross-check (Section 4.10.1) itself has weak, and on every re-measurement chance-level or worse, row-level agreement with the LLM judge (Pearson $r = -0.300$ as of the 2026-08-02 re-measurement, down further from $r = -0.063$, down originally from $r = 0.089$) – the two instruments now also disagree on which configuration has the best raw faithfulness score, not just on individual rows; we no longer treat this cross-check as corroborating evidence in either direction (Section 4.10.1 explains why), and note that building a genuinely reliable independent faithfulness signal remains open – to that end we prepared, but have not yet completed, a 50-item disagreement-prioritized human annotation sample (the query_ids where the LLM judge and the NLI check disagree most, plus a smaller agreement-anchor subsample), with blank rating columns for a human rater; this closes the machine-side half of a future human validation pass but the actual annotation, requiring a person, remains outside this revision's scope. Every prior re-measurement of this cross-check kept the same NLI model (`cross-encoder/nli-MiniLM2-L6-H768`, ∼66M parameters, originally chosen for RAM reasons while a concurrent job used most of available memory) and only re-ran it against fresh data, never asking whether a stronger model changes the picture – a genuinely untried angle, and a purely instrumental one, since it touches only how already-generated answers are re-scored after the fact, not any retrieval, generation, or abstention behavior. Re-running the identical SummaC-ZS methodology on the identical 169-row sample with `cross-encoder/nli-deberta-v3-base` (∼370M parameters, same confirmed contradiction/entailment/neutral label ordering) finds a real, substantial improvement: Pearson r moves from -0.300 to -0.076, and binary agreement (both instruments above/below the 0.5 midpoint) moves from 0.556 – barely better than a coin flip – to 0.822. This does not rescue the cross-check as corroborating evidence: the correlation, while much closer to zero, is still not positive, so we still do not treat it as confirming the LLM judge. But it substantially revises the earlier diagnosis: the near-chance row-level agreement was, at least in part, an artifact of the smaller model's limited capacity rather than proof that sentence-level NLI and LLM-as-judge faithfulness are fundamentally incompatible measurement approaches on this corpus – a materially more encouraging (if still not decisive) starting point for the human-validation pass this item already calls for. `scripts/measure_nli_faithfulness_deberta_roundP.py`; results in `results/nli_vs_llmjudge_agreement_roundP_deberta.csv`. A third model swap pushed this further still: `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`, fine-tuned not just on generic entailment data but on MultiNLI, Fever-NLI, ANLI, LingNLI, and WANLI combined (885k pairs) – Fever-NLI specifically reformulates FEVER (Thorne et al., 2018) as claim-vs-evidence NLI pairs, structurally closer to our own claim-vs-context faithfulness check than plain SNLI/MultiNLI entailment. On the identical 169-row sample, this moves Pearson r to -0.018 and binary agreement to 0.905 – the closest to zero and the highest agreement of all three models tested (MiniLM: $r = -0.300$, agreement 0.556; DeBERTa-base: $r = -0.076$, agreement 0.822; DeBERTa-large-FEVER: $r = -0.018$, agreement 0.905), a consistent, monotonic trend across three independent model swaps, though the correlation is still not positive. This third run also surfaced a genuine hardware-performance finding worth disclosing on its own terms: on CPU, this specific ∼435M-parameter model needed over 9 hours and completed only 8 of 38 batches (the first batch alone took 3h08m) before we terminated it as impractical – a disproportionate slowdown relative to model size, not explained by memory pressure (VRAM/RAM headroom was not the bottleneck). Re-run on the local GPU (NVIDIA RTX 3050 Ti, 4GB VRAM, confirmed idle and available first) instead, it completed the identical computation in 18 seconds. `scripts/measure_nli_faithfulness_fever_roundP.py`; results in `results/nli_vs_llmjudge_agreement_roundP_fever.csv`. A fourth attempt, using a model trained directly for document-claim grounding verification rather than repurposed generic NLI (MiniCheck, Tang et al., 2024), did not produce a result, for a disclosed reason rather than a silent gap: its only published checkpoint is a legacy pickle-format file with no safetensors variant, and current `transformers` refuses to deserialize non-safetensors checkpoints without a newer torch version than this environment has installed, a deliberate block against a real `torch.load` deserialization vulnerability (CVE-2025-32434). We did not upgrade torch or bypass this check mid-session to force the experiment through, since either risks the reproducibility of every other already-verified component or knowingly circumvents a security control for an untrusted pickle file – neither is a call to make casually to finish one more experiment. Eighth, the FacultyAvailability coverage gap identified in an earlier pass has been closed (Section 4.8), and we generalized the underlying lesson into a standing table-coverage audit (Section 3.10) rather than leaving it as two one-off fixes; the audit currently finds no remaining zero-coverage table, though it can only ever prove a table has *some* exercising query, not that its routing is correct in every case. Ninth (resolved, corrected here): an earlier revision of this list described the LLM-based entity-normalization fallback (Section 3.7) as not yet validated by its own ablation and disabled by default – this was already inaccurate at the time relative to the actual deployed code (`use_entity_normalization` defaults to `True`, validated 2026-07-28 at 8/8 on a hand-constructed malformed-query set: misspelled names, phonetic errors, and course-code spacing/punctuation variants a fixed regex alone does not catch), and we correct the record here rather than let a stale disclosure stand. Re-running that exact ablation this revision, under the corrected corpus, reproduces the identical result (8/8 with normalization vs. 1/8 without, `scripts/eval_entity_normalization.py`) – a small n, disclosed as such, but a clean, fully reproduced signal, not a marginal one. Tenth, the expanded cross-lingual stress test (27 queries, three LLM-generated Banglish phrasings each of the same 9 underlying facts) shows a more cautious picture than the original 9-query set, and more cautious still on re-measurement: the direction still nominally favors query translation (59.3% vs. 63.0% top-5 accuracy as of 2026-08-01, both figures higher than previously measured because the underlying retriever itself improved) but is not statistically significant on the discordant pairs (exact binomial $p = 1.0$, up from $p = 0.219$), and manual inspection found visible naturalness problems in a minority of the higher-temperature-generated variants, consistent with Doğruöz et al. (2023)'s caution against treating LLM-generated code-switched text as equivalent to organically-collected data. The retriever-wide improvements made elsewhere this revision (BM25 retuning, the exact-match fix, the embedding retrain) have the side effect of narrowing exactly the gap this stress test targets, on both the original and expanded sets – a real, disclosed interaction between unrelated fixes and this specific evaluation, not a sign the fixes were unhelpful.

5.2 Future Work

The most direct remaining priority is a human hallucination annotation study on the current pipeline, with a second rater, to complement the automated faithfulness score. This is, if anything, more urgent than before: an independent, architecturally-separate NLI-based faithfulness cross-check (Section 4.10.1), re-measured this revision, no longer cleanly corroborates the claim-decomposition-artifact hypothesis the way the previous measurement did – row-level agreement with the LLM judge is now indistinguishable from chance, so neither confirms nor rules out that the LLM-judge gap is a genuine grounding regression. A human rater remains the direct way to settle this, and the automated cross-check's own instability under re-measurement is itself an argument for why an automated proxy alone should not be trusted to close this question. A second priority is extending the cross-lingual stress test's small sample (Section 4.11.4) with more EnglishQA-only content as the corpus grows, since its current 9-query size limits it to a suggestive rather than a confirmatory result, even though its direction (query translation roughly doubling retrieval accuracy in the scenario it targets) is now demonstrated rather than merely hypothesized. A third is re-testing the sufficient-context override's net effect on precision/recall now that the underlying confidence gate is substantially more sensitive (Table 3) than when the override was first validated.

This paper extended a hybrid BM25/dense RAG chatbot for academic advising into an adaptive pipeline combining domain-fine-tuned bilingual embeddings, per-query-type fusion routing, prerequisite-graph augmentation, and confidence-gated abstention with a sufficient-context override, and evaluated it on 200 English and 100 Banglish test queries with paired significance testing throughout, plus standard IR metrics (Recall@k, MRR, nDCG@10) and an independent NLI-based faithfulness cross-check. Under the current retriever and corpus, the adaptive pipeline is in full statistical parity with BM25-only on all four automated metrics, and remains statistically tied with full hybrid on all four as well; against vector-only the only close case (METEOR) is inconclusive, disagreeing between a paired t-test and a matched Wilcoxon test; a deconfounding diagnostic found the router's original fusion-method choice (RRF for entity-heavy queries) significantly negative in isolation once the exact-match mechanism's ranking ceiling was removed, so the observed parity with the fixed baselines was borrowed entirely from that separate, independently-validated mechanism rather than being evidence that choice was beneficial – we replaced it with linear fusion at the same weighting as a direct, verified consequence, preserving the tie exactly while narrowing a residual ranking-quality gap – a small significant remainder of that gap, root-caused to a genuine faculty near-duplicate-name confusion, was subsequently closed with a small, safety-verified same-entity ranking bonus (Section 4.9.2), with zero measured effect on Recall/MRR. The system's real, defensible value is the abstention and multi-hop graph reasoning capabilities the baselines cannot perform at all, together with the entity-collision handling that drives the exact-match mechanism's own positive result, not a generation-quality edge over the single-method baselines. Embedding fine-tuning and prerequisite-graph augmentation (confirmed on all four metrics, both significance tests, once measured on an expanded, still entirely real, $n = 31$ sample) each produced a measured positive effect; the unambiguous-match score guarantee (extended in this revision to cover faculty lookups, not just course codes) also produced a measured positive effect historically, but its marginal contribution on top of the additive exact-match bonus alone has since converged to zero on the current, further-improved retriever – still live and harmless in production, but no longer independently demonstrable on this test set; a cross-encoder reranker – both a generic version and, after we caught and corrected a training/deployment mismatch, a properly fine-tuned one – produced a measured negative effect in both cases, traced specifically to conflict with the exact-match mechanism on entity-heavy queries; restricting reranking to the open-ended route alone (a route-conditional extension of this paper's own adaptive-routing principle) resolves the significant loss into a clean tie – not a win, but the first tested configuration that does not measurably hurt; cross-lingual query translation showed a null effect on the main Banglish test set but a positive, now twice-replicated effect (61.5% to 84.6% top-5 accuracy at $n = 13$, narrower than first measured after this revision's other retriever fixes partly closed the same gap) on a dedicated stress test built for the scenario it targets, though this effect no longer reaches significance on a larger, noisier expanded version of the same stress test; and BM25-level code-mixed normalization remains a null result on the data tested. We describe the result as a validated system-level integration for an under-studied bilingual advising setting, disclose every limitation and every methodological pitfall (including our own) found along the way, and identify human hallucination annotation as the most direct remaining step toward a stronger generalization claim.

ETHICAL STATEMENT

To protect student privacy, personal identifiers (names and student IDs) were removed from the dataset during ingestion, and community-sourced data was anonymized prior to analysis.

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- **Financial and other Support:** No funding was received for the work reported here.
- **Author Contributions:** The authors contributed equally.
- **Data Sharing:** A dataset was generated and analyzed during the current study, available at <url>.
- **AI Tools:** The paper used AI tools for text correction purposes only.

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